Curvature-balanced learning rates

Each layer a (embeddings; transformer blocks 1-n; final norm + head) trains at its own rate eta_a(t) = eta(t) g m_a, with eta(t) the tuned schedule of the baseline (warmup, then cosine to 10%). The balanced operator is Hhat = D^1/2 H D^1/2, D = diag(eta_a) for SGD and diag(eta_a) P^-1 for AdamW (P frozen from the current second moments), with H the Hessian on 32 fixed training sequences. Its linear-stability threshold is 2 + 2 mu = 3.8 for SGD (heavy ball) and (2 + 2 beta1)/(1 - beta1) = 38 for AdamW. With v the top eigenvector of Hhat, the share of layer a is s_a = ||v_a||^2 = d log lambda_max / d log m_a (sum_a s_a = 1); equal shares maximize prod_a m_a at fixed lambda_max. Every 20 steps a warm-started block iteration (3 vectors, 2 Rayleigh-Ritz steps on the vectors and their residuals) refreshes the top eigenpairs; shares are averaged over the Ritz vectors within 5% of the top, then smoothed (EMA weight 0.2). A 20-step Lanczos check of lambda_max(Hhat) runs every 500 steps. Multipliers stay at 1 during warmup.

Muon (orthogonalized updates) is balanced on its spectral-norm sharpness instead (Islamov et al.): with c_l = m_a times Muon's shape scale for each hidden matrix l, S = max <D, H D> over updates with every ||D_l||_2 <= c_l, and the stability ratio is rho = eta S / (2 sum_l c_l ||grad_l||_*) (edge at 1). S is tracked by 5 warm-started Frank-Wolfe steps per refresh (checked from scratch every 500 steps); the share of layer a is d log rho / d log m_a = 2 <D_a, (HD)_a> / <D, HD> minus layer a's share of the c-weighted nuclear norm (sum 1). Embeddings and head (AdamW in Muon) keep the schedule's rate.

Variants

A (redistribute):	log m_a -= alpha (s_a - 1/L), recentred to geometric mean 1; g = 1; m_a in [0.1, 10].
B (A + scale):	also log g -= gain log(lambda_max(Hhat) / (target x threshold)); every g m_a in [0.1, 10]. B<t> targets t x threshold (Muon: rho = t).
-fe	embedding kept at the schedule's rate (still in the operator): the 32 sequences see only part of the vocabulary, and SGD B0.9 blew up with the embedding at 10x.
-fresh	Hessian on 32 fresh training sequences at every refresh instead of the fixed ones.
plain-lr<x>	unbalanced control at base rate x.

Defaults: alpha = 0.3, gain = 0.2 per refresh. AdamW A-fe tracks 4 vectors with 3 iterations. Batch sharpness is not measured in these runs (memory).

v2 (AdamW and Muon; runs named R)

Every 20 steps, along the step s the optimizer just took (without weight decay), on 256 fresh training sequences (gradient g and Hessian H at the parameters before the step): R_a = s_a^T (H s)_a / (-2 g_a^T s_a) for each layer (the sums over layers give the whole step's R, which sits near 1 for AdamW and Muon at their tuned rates). EMA (0.2) of numerators and denominators, Rbar_a = their ratio; after warmup log m_a -= 0.3 (log Rbar_a - mean), the deviation clipped to [log 0.1, log 10] (a layer whose step does not lower the loss to first order counts as too fast); geometric mean of m fixed at 1, m_a in [0.1, 10]; no global scale. Every layer, Muon's AdamW part (embedding, head) included. The old measure (shares on the 32 fixed sequences) is logged alongside. Ro: the layer's own block s_a^T H_aa s_a in the numerator instead (no cross-layer terms).

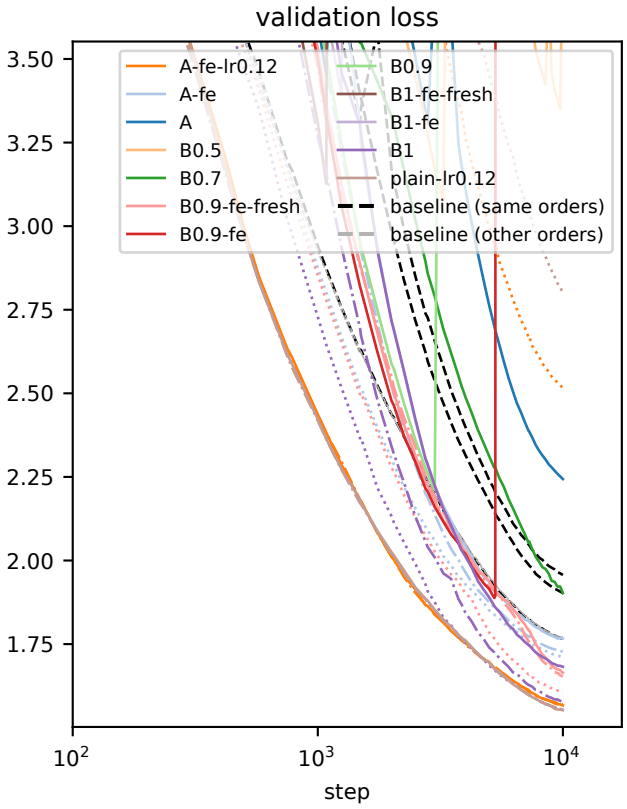
Finding: R_a settles at the same value in every layer (about 1; about 0.3 for the own block) whatever the layer's rate, even with rates 100x apart, so each layer sits at its own edge and the rule random-walks the rates into the caps.

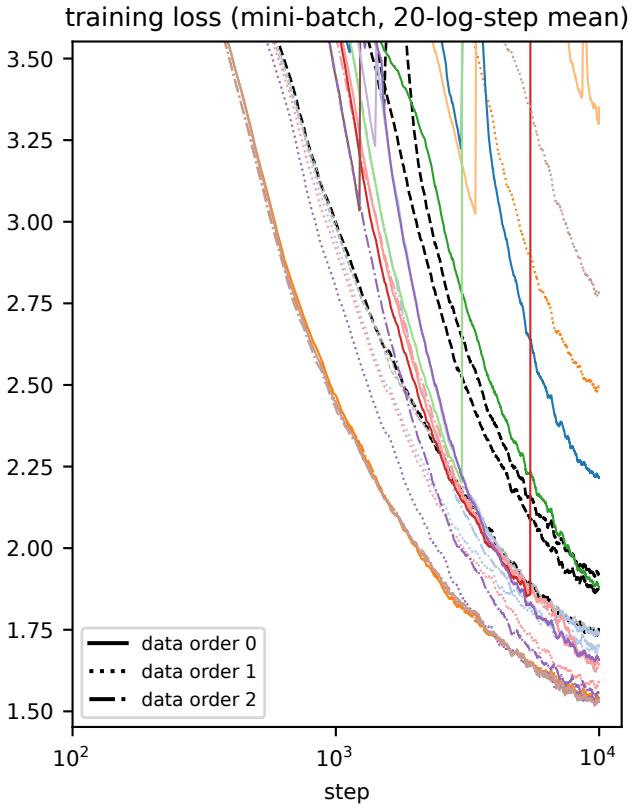
Comparisons

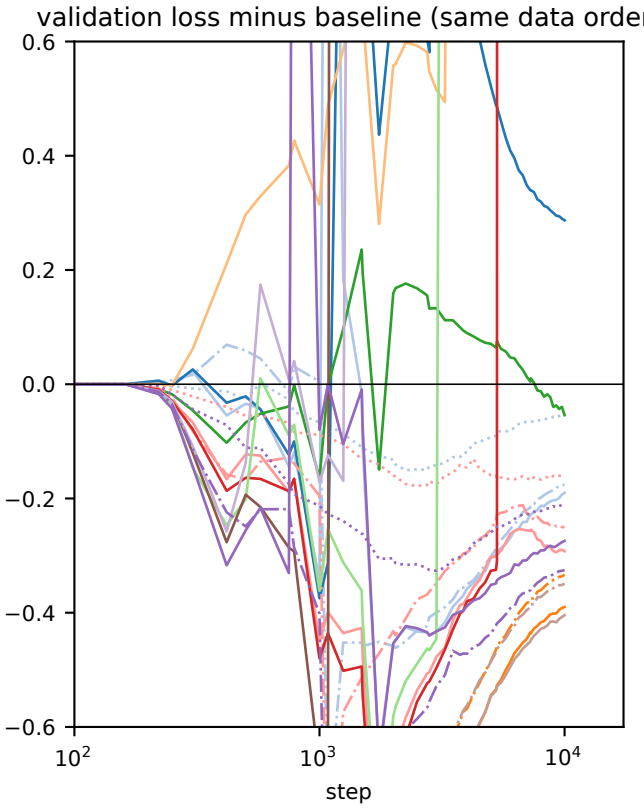
Baseline: the main runs (same initialization, same rate, same data orders). Speedup = the baseline's step at its best validation loss / the first evaluated step at which the balanced run reaches it (same data order); the best validation loss is the final one except on Shakespeare, where the runs overfit.

JS: Jensen-Shannon divergence of next-token distributions on the fixed validation contexts, at equal training loss, between a balanced run and the baseline on the same data order; the grey band is the baseline's floor (min to max over pairs of its data orders).

v1 SGD, TinyStories-10M: losses

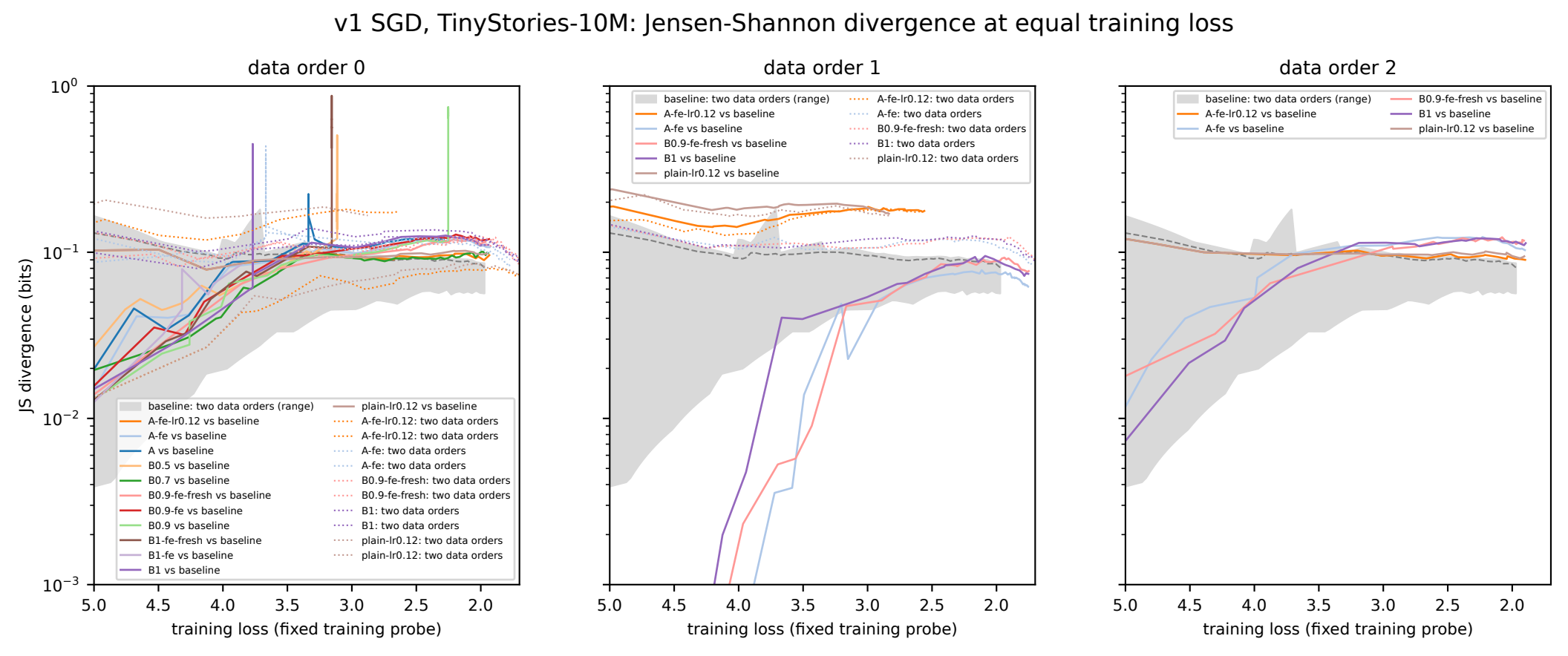




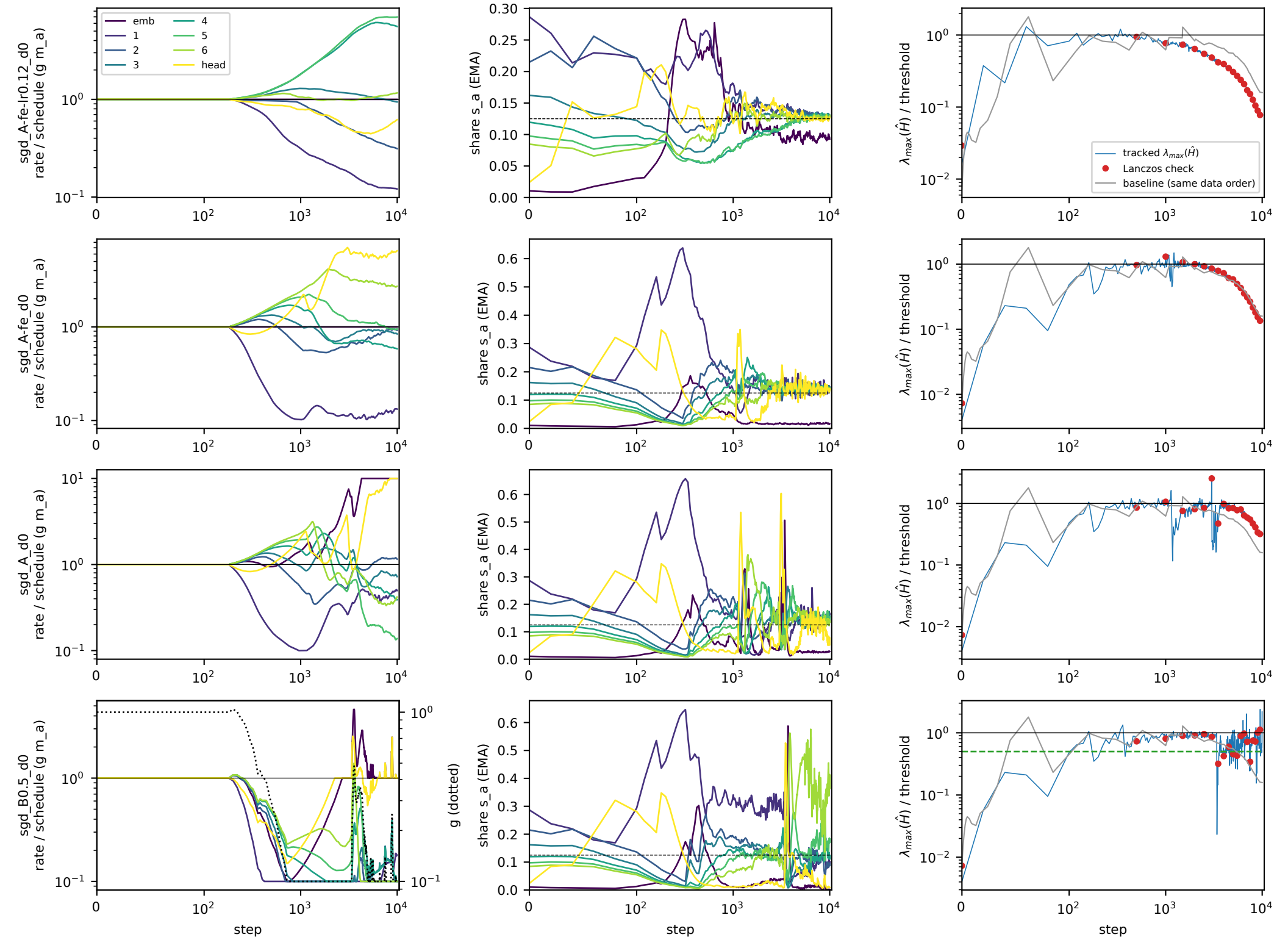


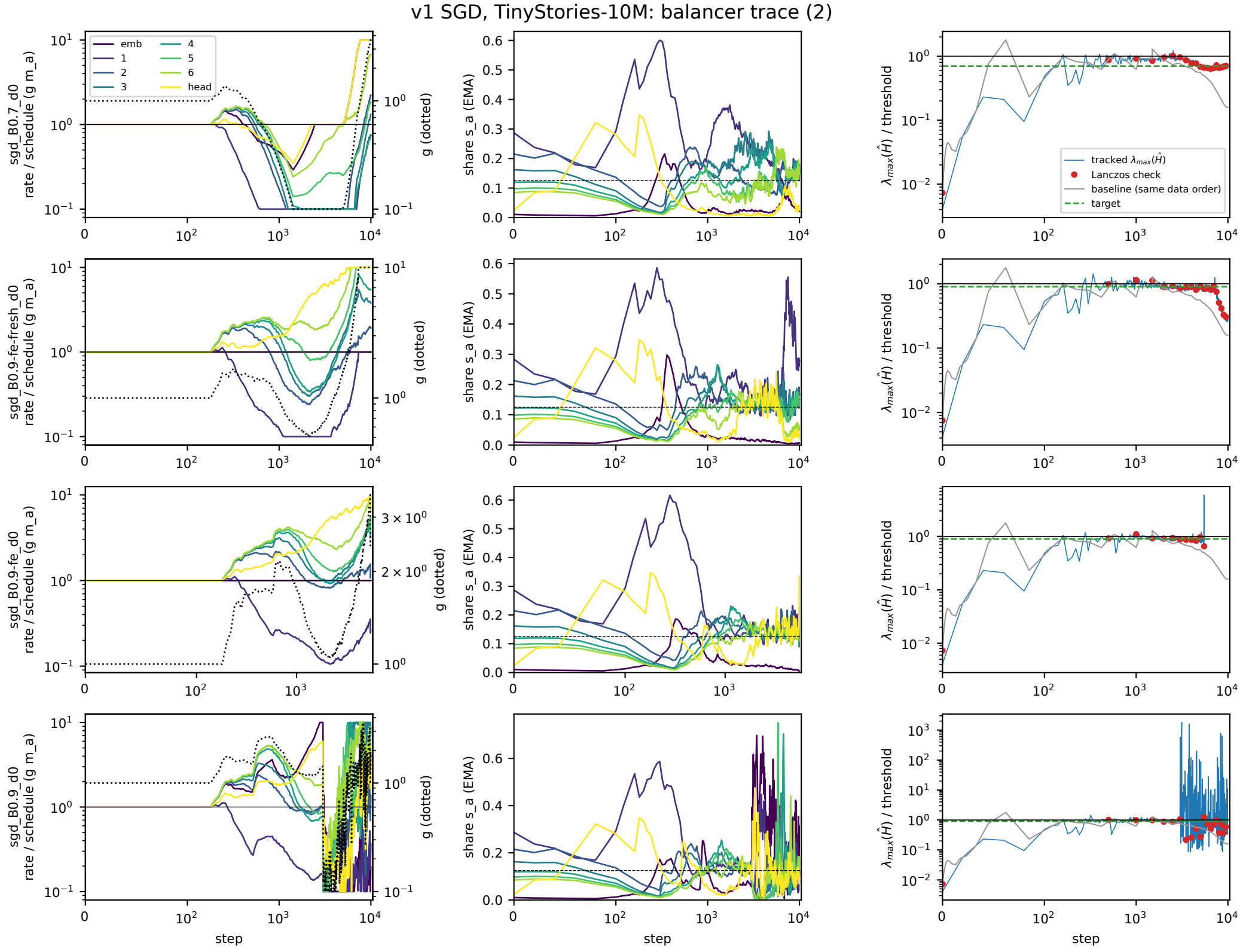
v1 SGD, TinyStories-10M: final and best validation loss

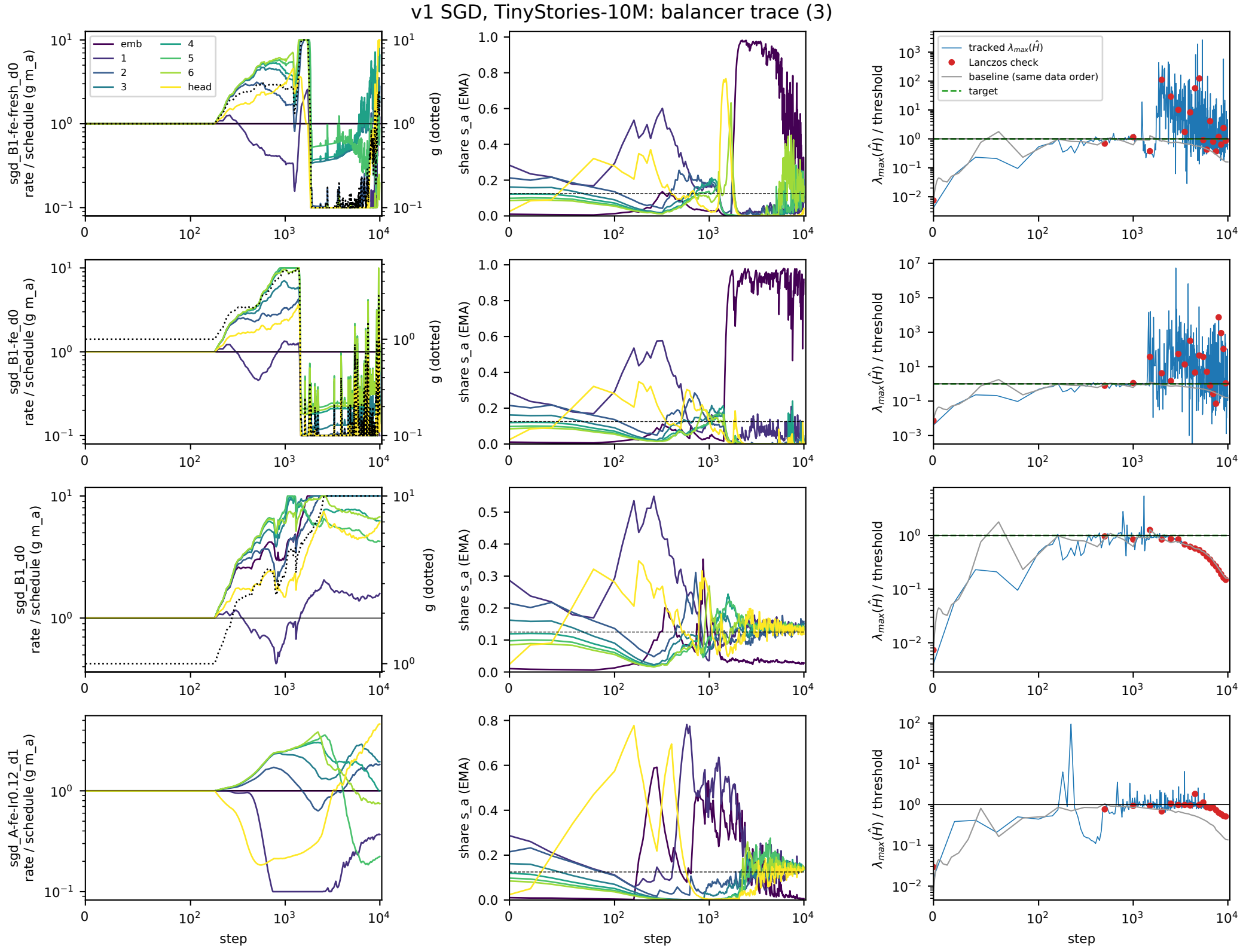
run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
sgd_A-fe-lr0.12_d0	10000	1.5666	-0.3900	1.5666	-0.3900	2250	4.44x
sgd_A-fe_d0	10000	1.7664	-0.1902	1.7663	-0.1903	4750	2.11x
sgd_A_d0	10000	2.2435	+0.2869	2.2435	+0.2869	-	-
sgd_B0.5_d0	10000	3.9069	+1.9503	3.0746	+1.1180	-	-
sgd_B0.7_d0	10000	1.9024	-0.0542	1.9024	-0.0542	9000	1.11x
sgd_B0.9-fe-fresh_d0	10000	1.6641	-0.2925	1.6641	-0.2925	5000	2.00x
sgd_B0.9-fe_d0	5500	38.2695	+36.3129	1.8878	-0.0688	4500	2.22x
sgd_B0.9_d0	10000	5.1156	+3.1590	2.2110	+0.2544	-	-
sgd_B1-fe-fresh_d0	10000	4.8535	+2.8969	3.1267	+1.1701	-	-
sgd_B1-fe_d0	10000	5.8562	+3.8996	3.3927	+1.4361	-	-
sgd_B1_d0	10000	1.6822	-0.2744	1.6822	-0.2744	4500	2.22x
sgd_plain-lr0.12_d0	10000	1.5522	-0.4044	1.5522	-0.4044	2500	4.00x
sgd_A-fe-lr0.12_d1	10000	2.5155	+0.7497	2.5155	+0.7497	-	-
sgd_A-fe_d1	10000	1.7094	-0.0565	1.7094	-0.0565	7250	1.38x
sgd_B0.9-fe-fresh_d1	10000	1.6040	-0.1619	1.6040	-0.1619	5298	1.89x
sgd_B1_d1	10000	1.5533	-0.2125	1.5533	-0.2125	3857	2.59x
sgd_plain-lr0.12_d1	10000	2.8009	+1.0351	2.8009	+1.0351	-	-
sgd_A-fe-lr0.12_d2	10000	1.5693	-0.3334	1.5693	-0.3334	2750	3.64x
sgd_A-fe_d2	10000	1.7274	-0.1753	1.7274	-0.1753	4750	2.11x
sgd_B0.9-fe-fresh_d2	10000	1.6530	-0.2497	1.6530	-0.2497	5500	1.82x
sgd_B1_d2	10000	1.5772	-0.3255	1.5772	-0.3255	3750	2.67x
sgd_plain-lr0.12_d2	10000	1.5537	-0.3490	1.5537	-0.3490	2750	3.64x

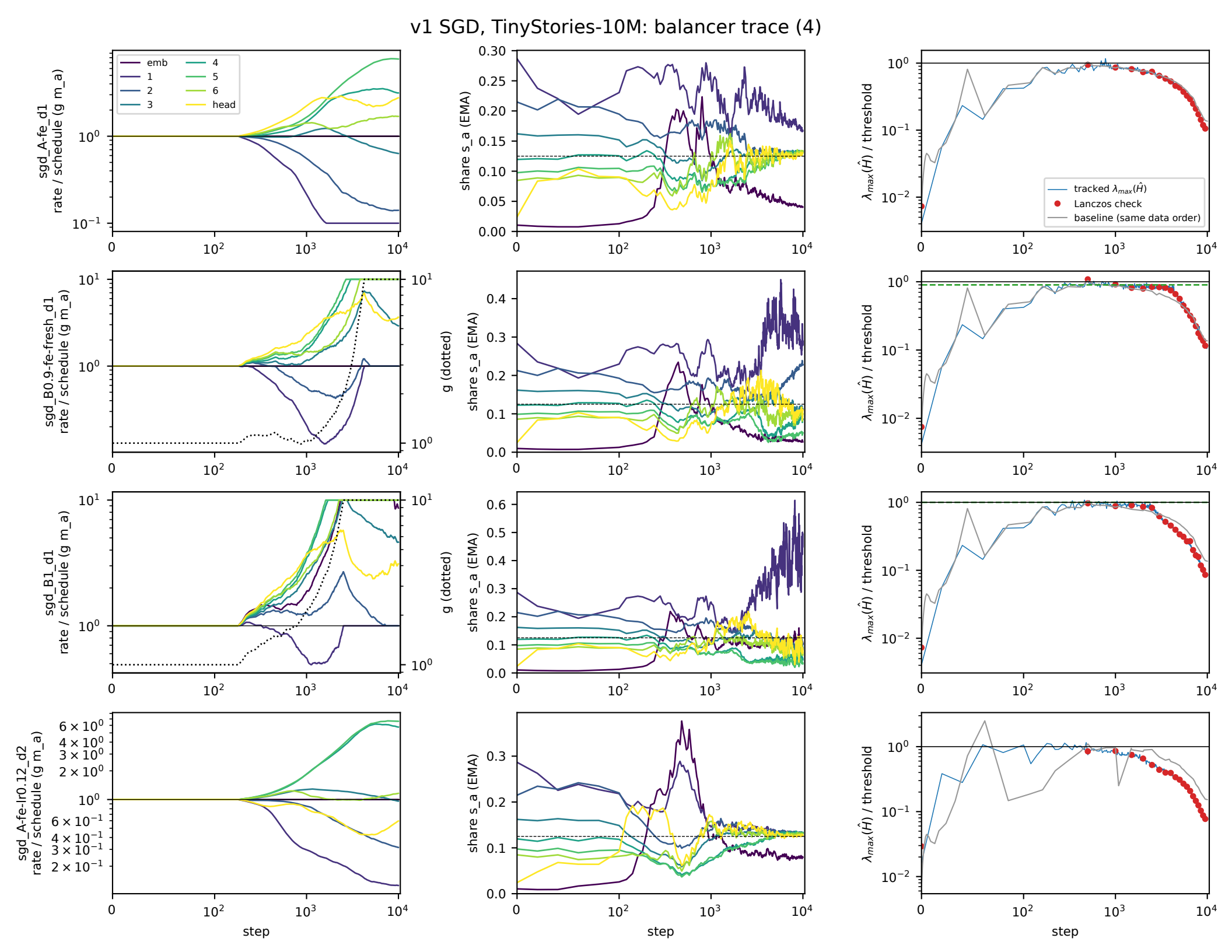

v1 SGD, TinyStories-10M: Jensen-Shannon divergence at equal training loss
data order 0
data order 1
data order 2
JS divergence (bits)
training loss (fixed training probe)
baseline: two data orders (range)
A-fe-lr0.12 vs baseline
A-fe vs baseline
A vs baseline
B0.5 vs baseline
B0.7 vs baseline
B0.9-fe-fresh vs baseline
B0.9-fe vs baseline
B0.9 vs baseline
B1-fe-fresh vs baseline
B1-fe vs baseline
B1 vs baseline
plain-lr0.12 vs baseline
A-fe-lr0.12: two data orders
A-fe: two data orders
B0.9-fe-fresh: two data orders
B1: two data orders
plain-lr0.12: two data orders

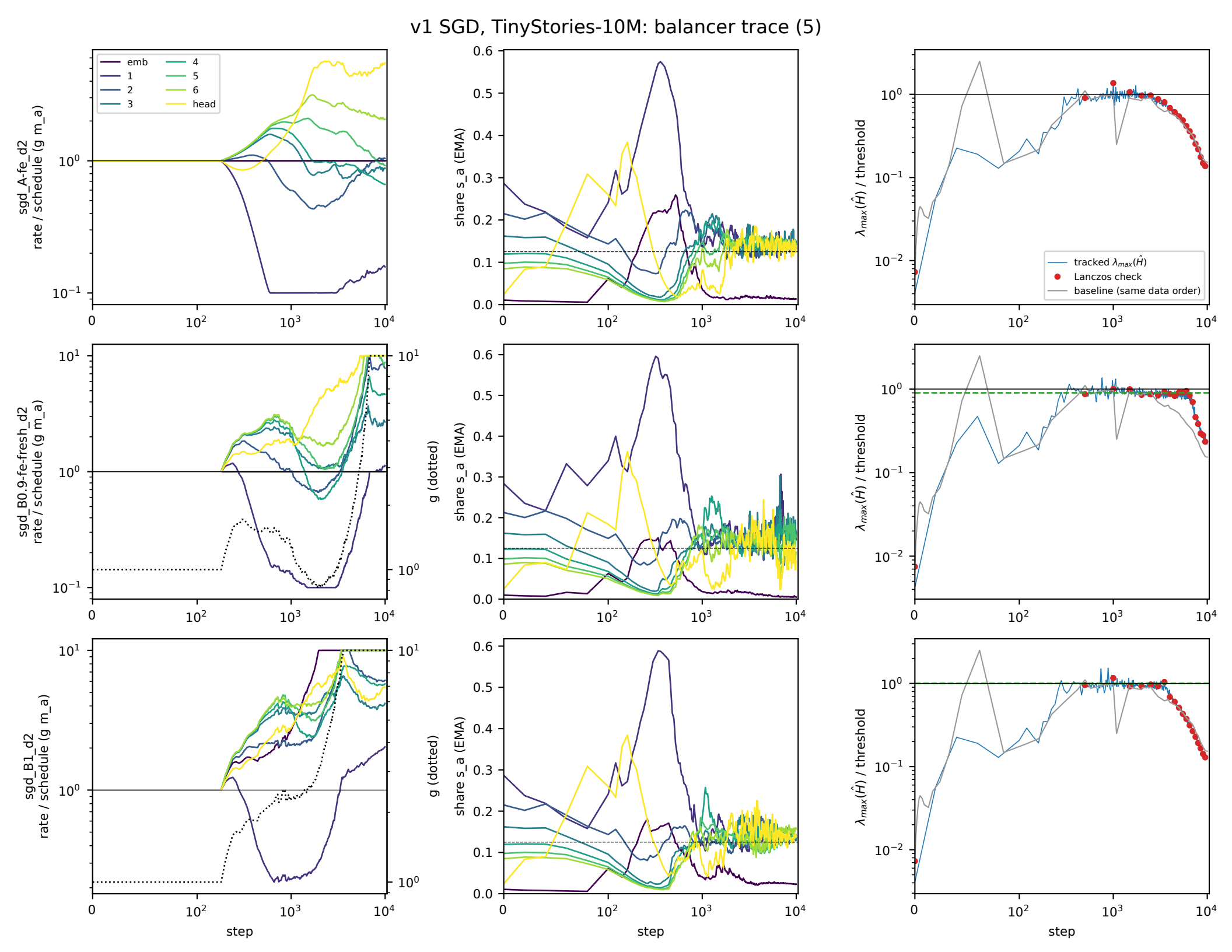
v1 SGD, TinyStories-10M: balancer trace (1)



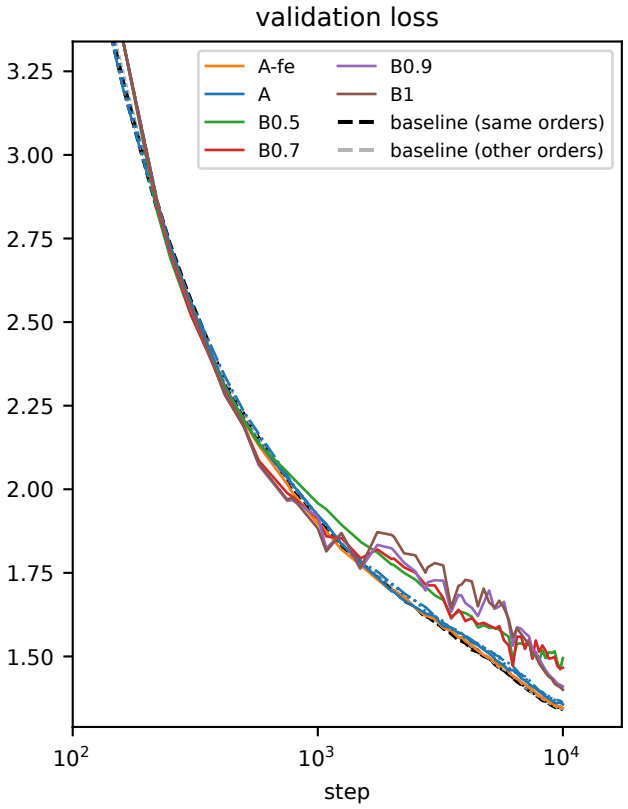

v1 SGD, TinyStories-10M: balancer trace (2)
sgd_B0.7_d0
sgd_B0.9-fe-fresh_d0
sgd_B0.9-fe_d0
sgd_B0.9_d0
rate / schedule (g m_a)
g (dotted)
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
emb
1
2
3
4
5
6
head
tracked $\lambda_{max}(\hat{H})$
Lanczos check
baseline (same data order)
target
step

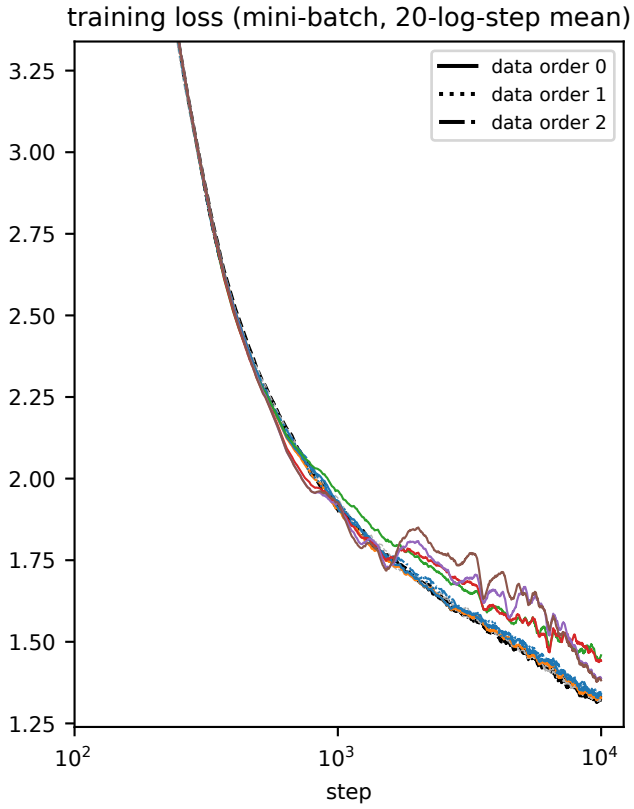

v1 SGD, TinyStories-10M: balancer trace (3)
sgd_B1-fe-fresh_d0
sgd_B1-fe_d0
sgd_B1_d0
sgd_A-fe-lr0.12_d1
rate / schedule (g m_a)
g (dotted)
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
emb
1
2
3
4
5
6
head
tracked $\lambda_{max}(\hat{H})$
Lanczos check
baseline (same data order)
target
step

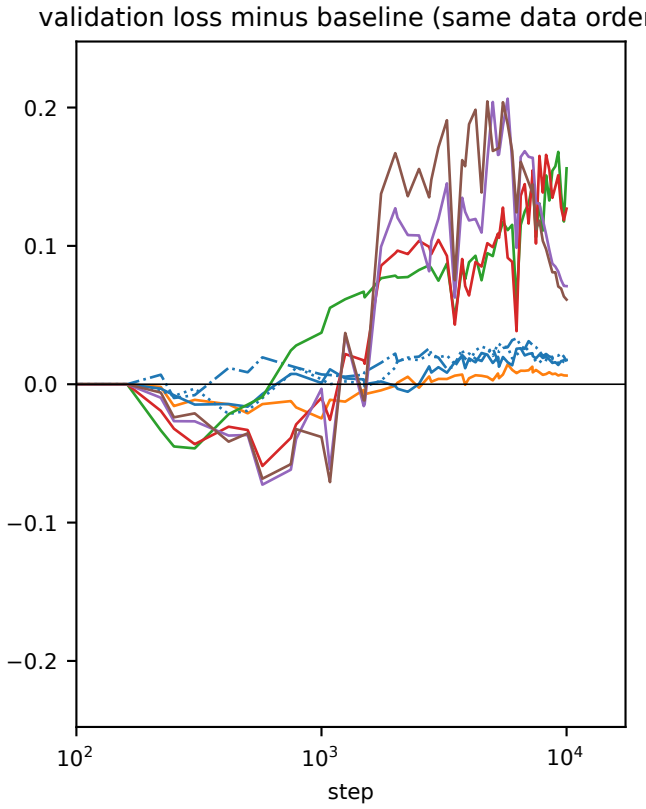

v1 SGD, TinyStories-10M: balancer trace (4)
sgd_A-fe_d1
sgd_B0.9-fe-fresh_d1
sgd_B1_d1
sgd_A-fe-lr0.12_d2
rate / schedule (g m_a)
share s_a (EMA)
g (dotted)
$\lambda_{max}(\hat{H})$ / threshold
emb
1
2
3
4
5
6
head
tracked $\lambda_{max}(\hat{H})$
Lanczos check
baseline (same data order)
step


v1 SGD, TinyStories-10M: balancer trace (5)
sgd_A-fe_d2
rate / schedule (g m_a)
emb
1
2
3
4
5
6
head
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
tracked $\lambda_{max}(\hat{H})$
Lanczos check
baseline (same data order)
sgd_B0.9-fe-fresh_d2
rate / schedule (g m_a)
g (dotted)
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
sgd_B1_d2
rate / schedule (g m_a)
g (dotted)
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
step
step
step

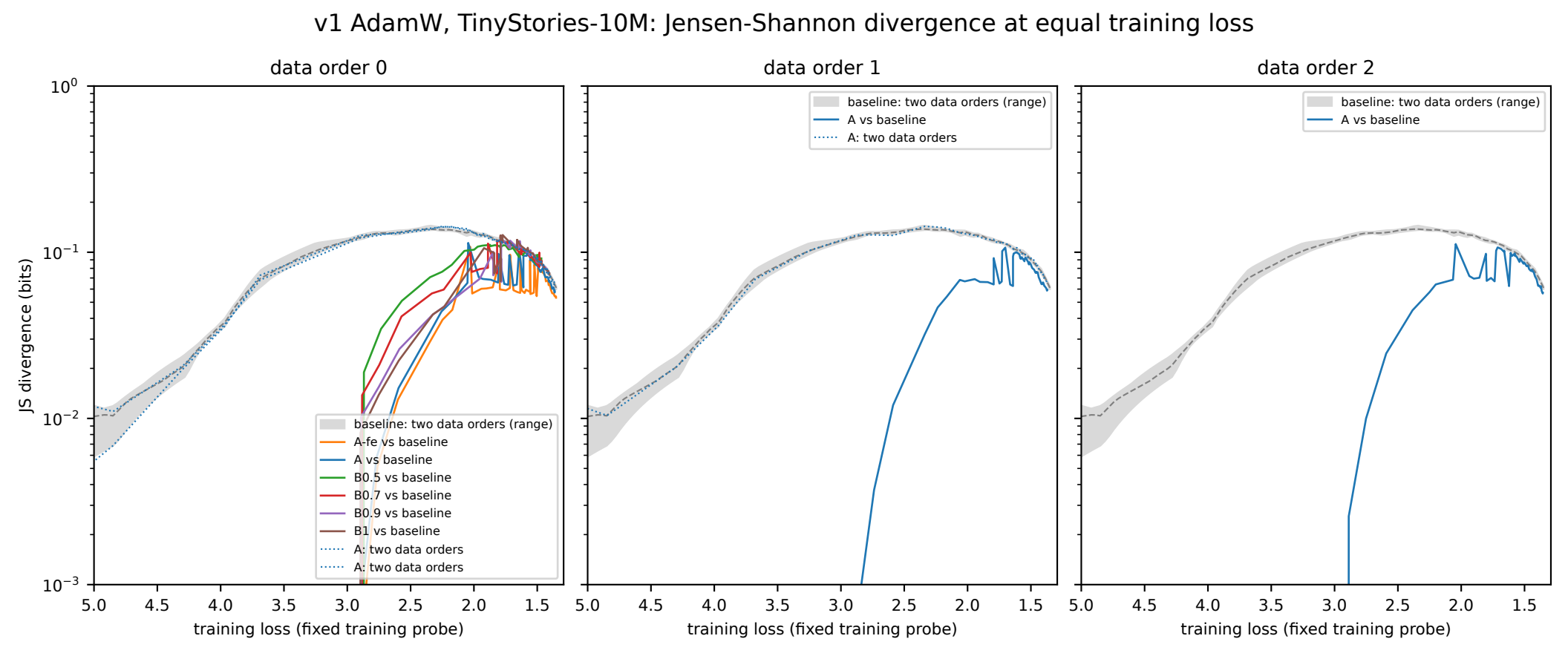
v1 AdamW, TinyStories-10M: losses

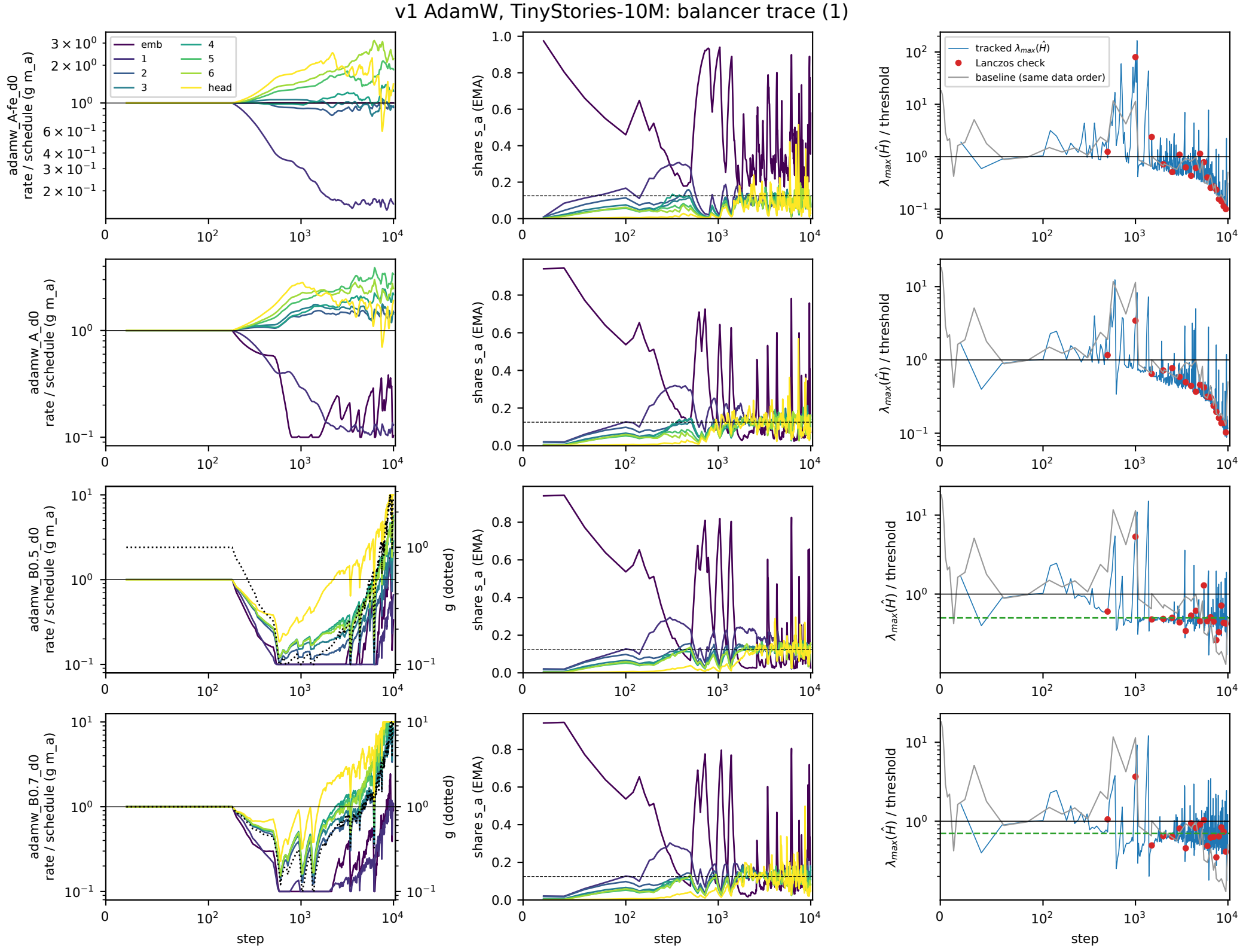




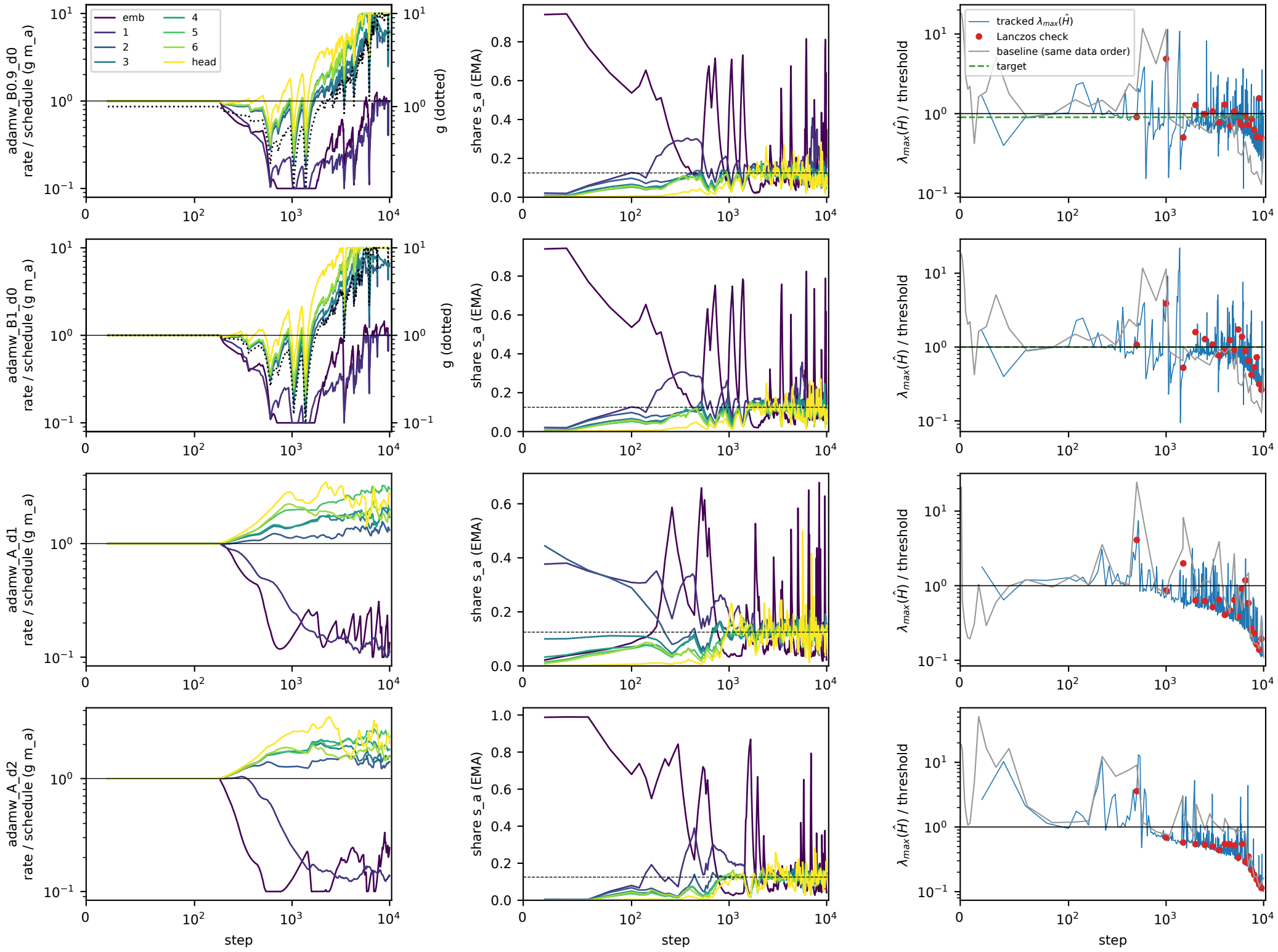


run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
adamw_A-fe_d0	10000	1.3454	+0.0063	1.3454	+0.0063	-	-
adamw_A_d0	10000	1.3567	+0.0175	1.3567	+0.0175	-	-
adamw_B0.5_d0	10000	1.4952	+0.1561	1.4618	+0.1227	-	-
adamw_B0.7_d0	10000	1.4660	+0.1269	1.4628	+0.1237	-	-
adamw_B0.9_d0	10000	1.4100	+0.0709	1.4100	+0.0709	-	-
adamw_B1_d0	10000	1.4003	+0.0612	1.4003	+0.0612	-	-
adamw_A_d1	10000	1.3615	+0.0153	1.3615	+0.0153	-	-
adamw_A_d2	10000	1.3611	+0.0165	1.3611	+0.0165	-	-

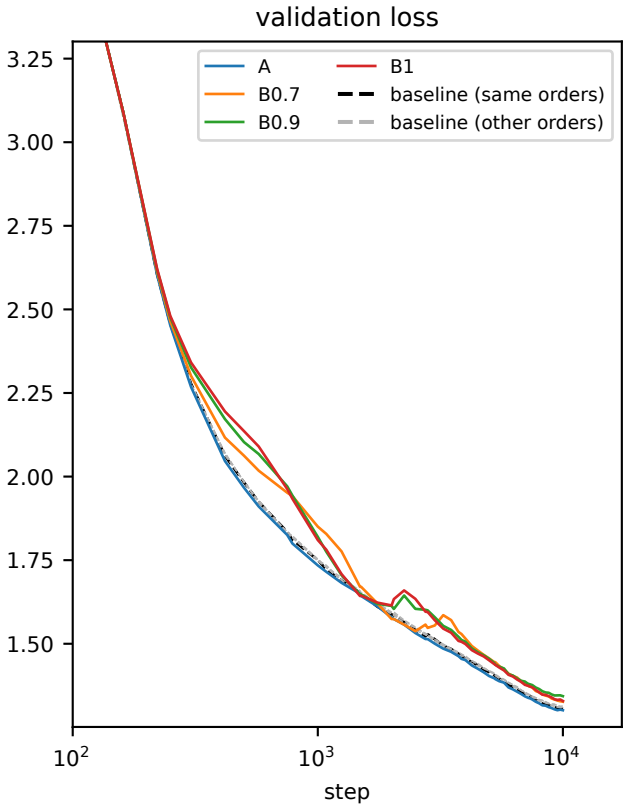

v1 AdamW, TinyStories-10M: Jensen-Shannon divergence at equal training loss
data order 0
data order 1
data order 2
JS divergence (bits)
training loss (fixed training probe)
baseline: two data orders (range)
A-fe vs baseline
A vs baseline
B0.5 vs baseline
B0.7 vs baseline
B0.9 vs baseline
B1 vs baseline
A: two data orders
A: two data orders

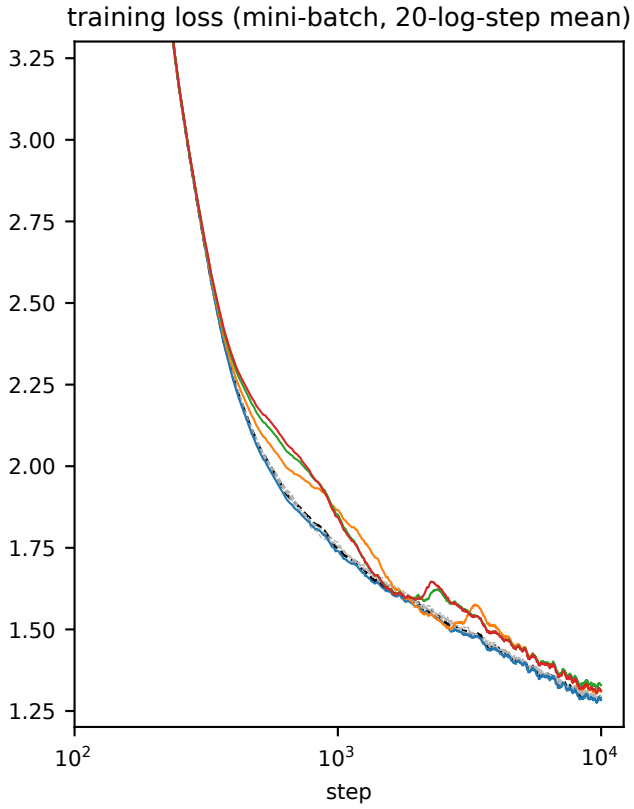

v1 AdamW, TinyStories-10M: balancer trace (1)
adamw_A-fe_d0
rate / schedule (g m_a)
adamw_A_d0
rate / schedule (g m_a)
adamw_B0.5_d0
rate / schedule (g m_a)
adamw_B0.7_d0
rate / schedule (g m_a)
emb
1
2
3
4
5
6
head
share s_a (EMA)
g (dotted)
tracked $\lambda_{max}(\hat{H})$
Lanczos check
baseline (same data order)
$\lambda_{max}(\hat{H})$ / threshold
step

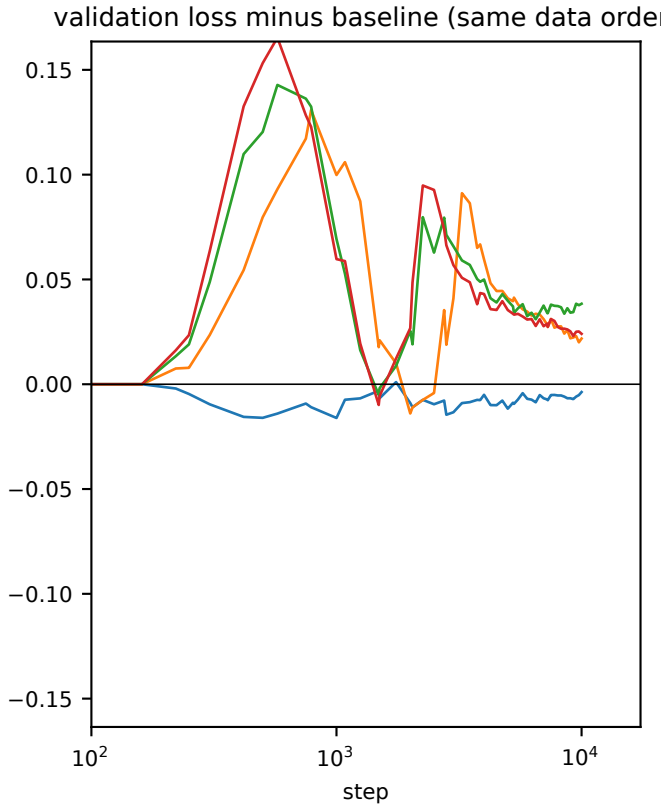
v1 AdamW, TinyStories-10M: balancer trace (2)



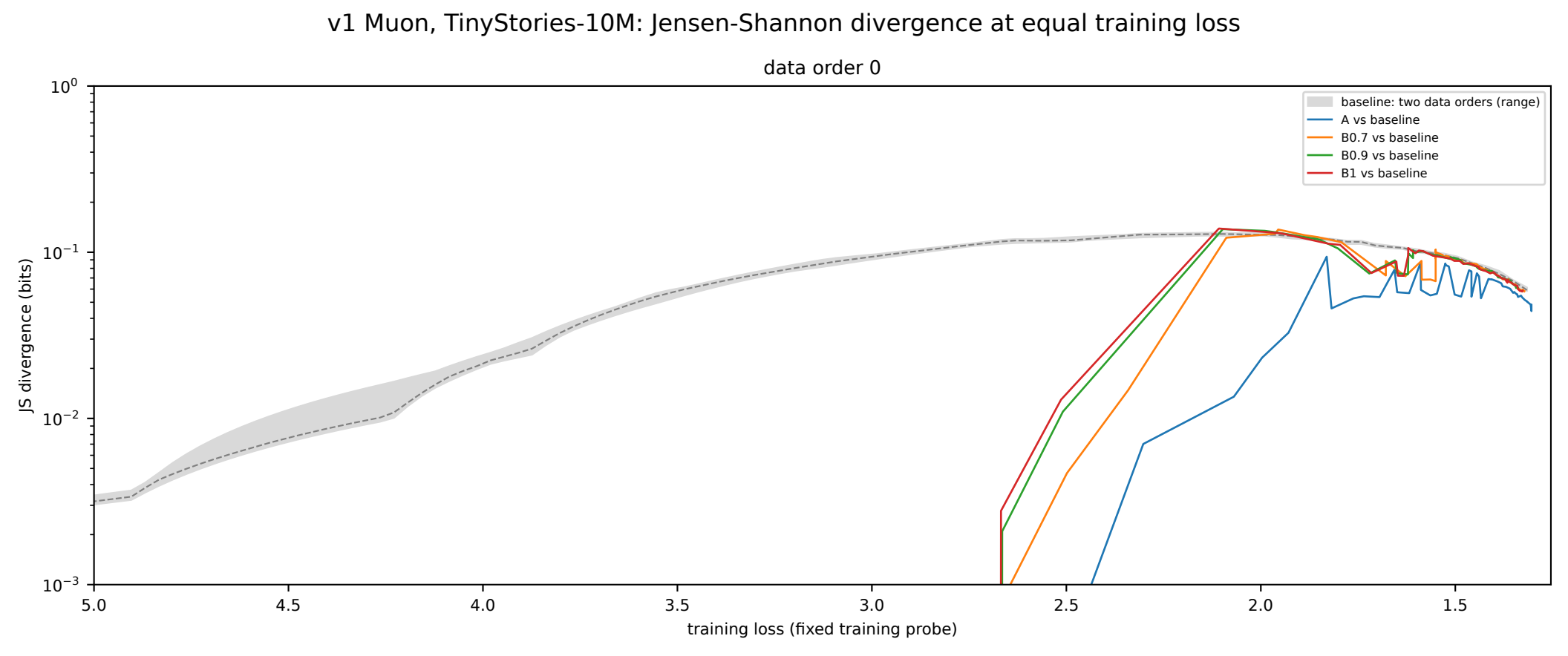
v1 Muon, TinyStories-10M: losses

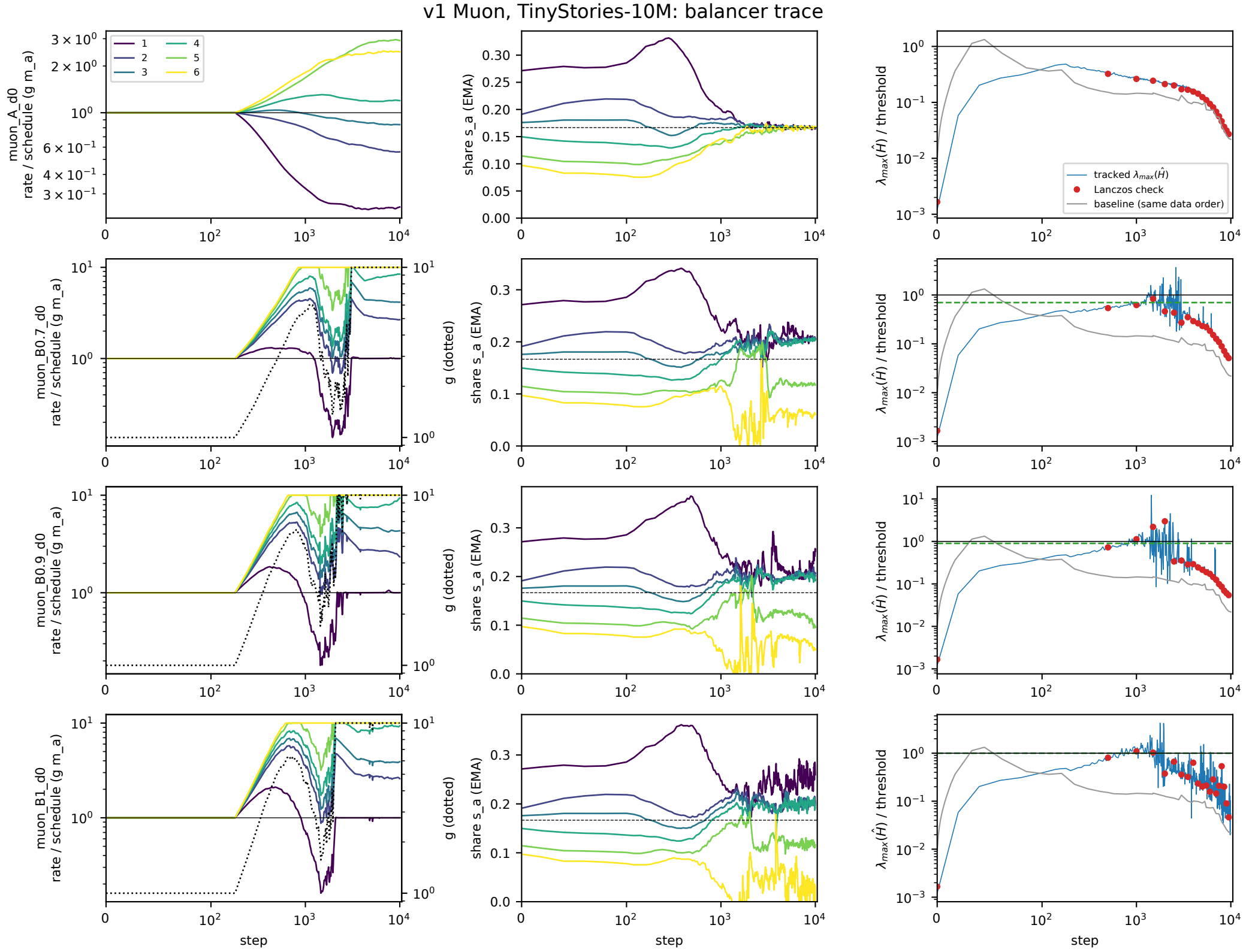




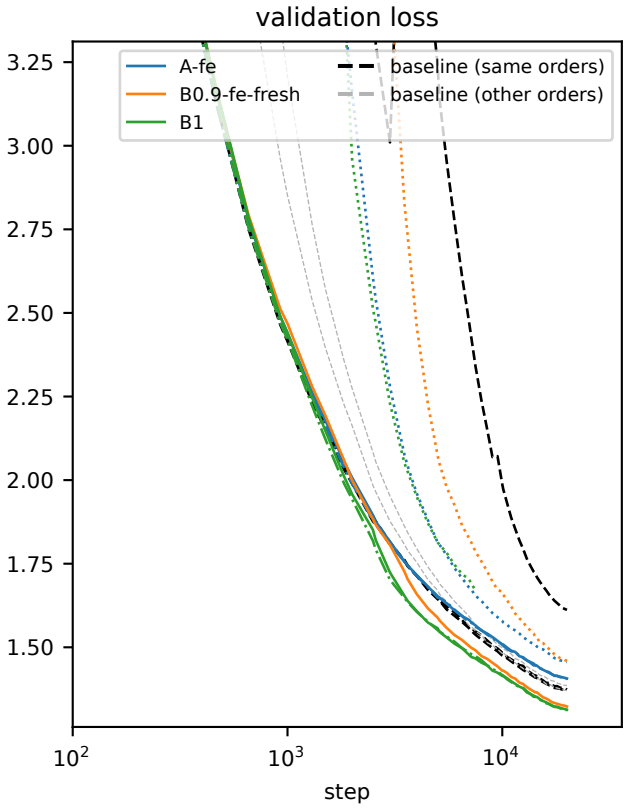


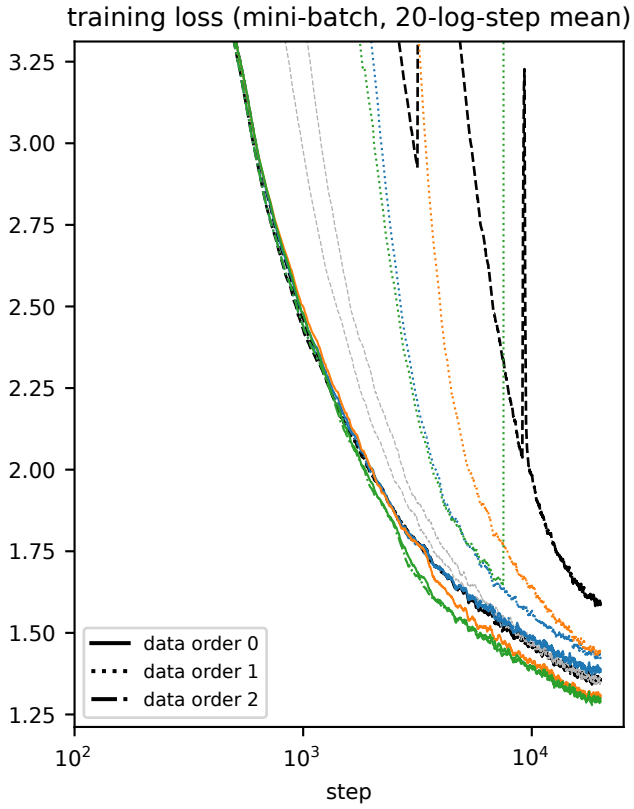
run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
muon_A_d0	10000	1.3012	-0.0037	1.3012	-0.0037	9250	1.08x
muon_B0.7_d0	10000	1.3267	+0.0218	1.3267	+0.0218	-	-
muon_B0.9_d0	10000	1.3433	+0.0384	1.3433	+0.0384	-	-
muon_B1_d0	10000	1.3290	+0.0241	1.3290	+0.0241	-	-

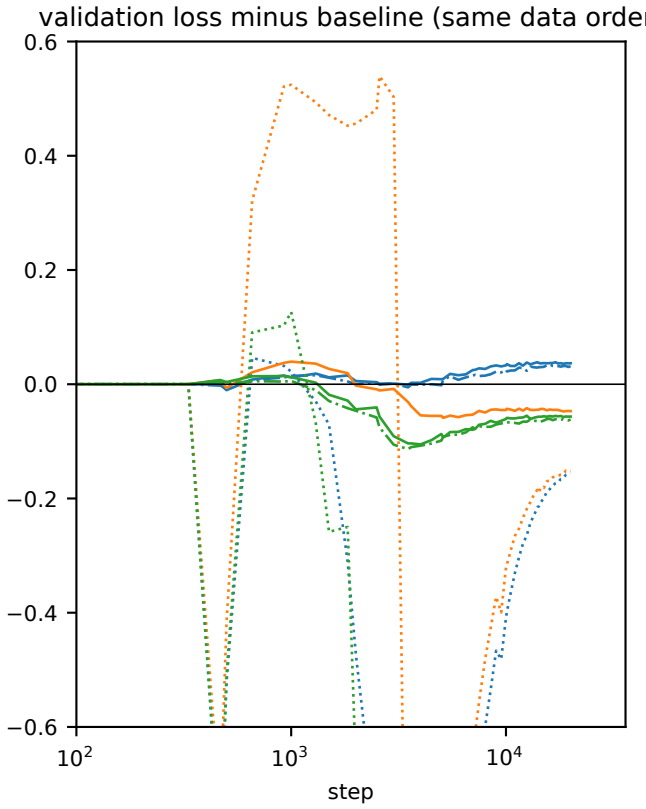

v1 Muon, TinyStories-10M: Jensen-Shannon divergence at equal training loss
data order 0
JS divergence (bits)
training loss (fixed training probe)
baseline: two data orders (range)
A vs baseline
B0.7 vs baseline
B0.9 vs baseline
B1 vs baseline


v1 Muon, TinyStories-10M: balancer trace
muon_A_d0
muon_B0.7_d0
muon_B0.9_d0
muon_B1_d0
rate / schedule (g m_a)
share s_a (EMA)
g (dotted)
$\lambda_{max}(\hat{H})$ / threshold
step
1
2
3
4
5
6
tracked $\lambda_{max}(\hat{H})$
Lanczos check
baseline (same data order)

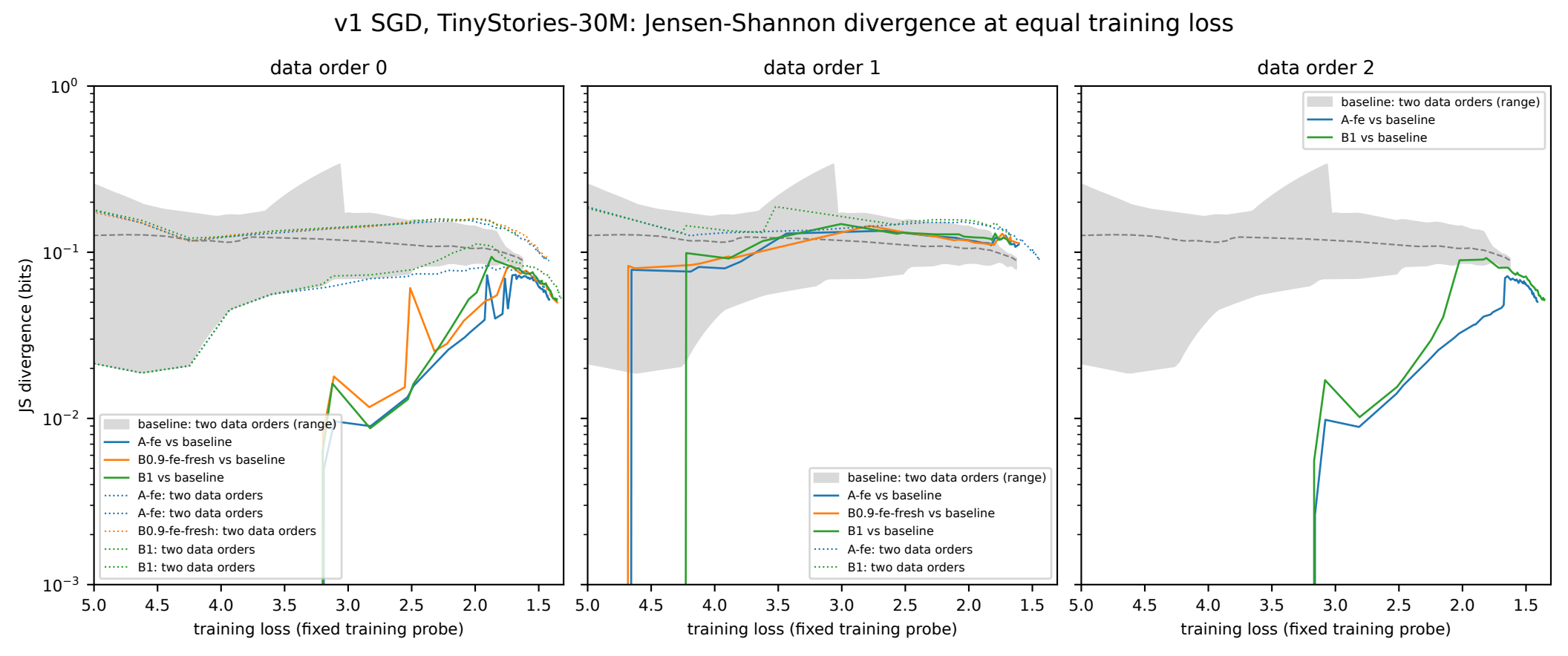
v1 SGD, TinyStories-30M: losses

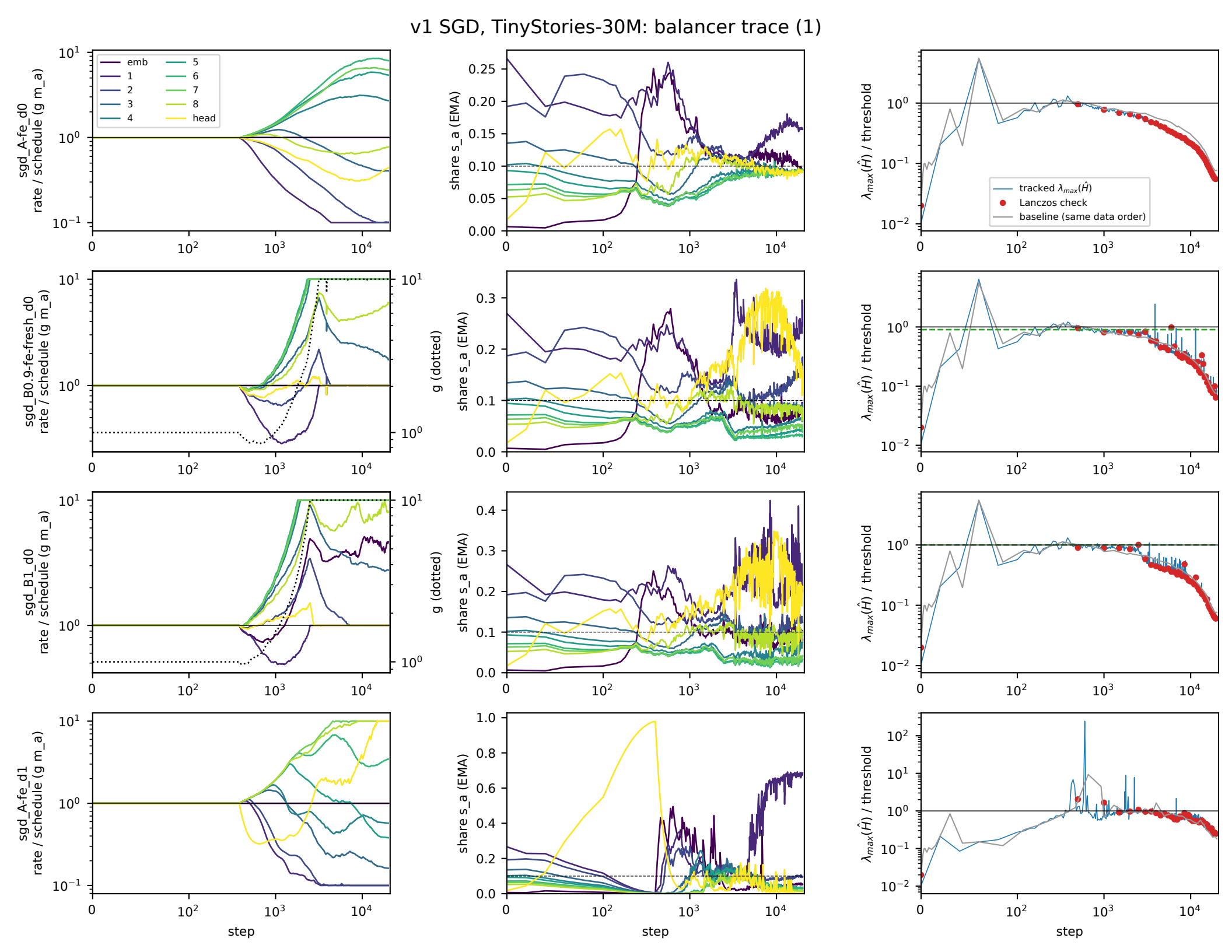


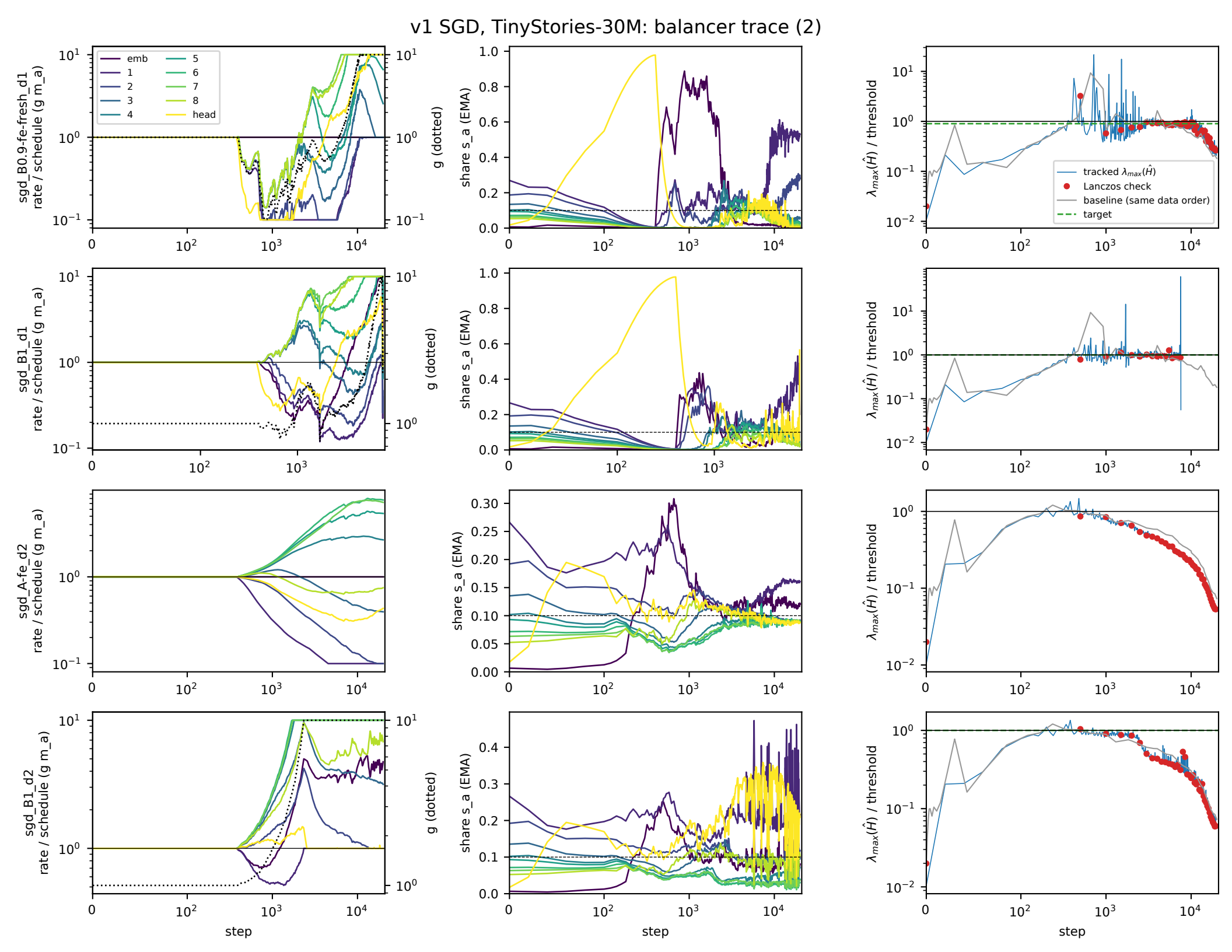




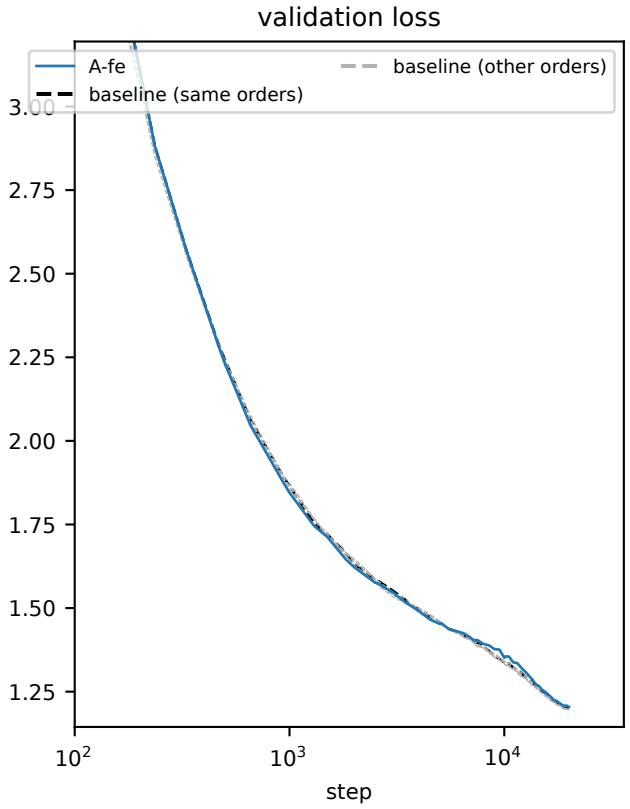
run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
sgd_A-fe_d0	20000	1.4066	+0.0367	1.4066	+0.0367	-	-
sgd_B0.9-fe-fresh_d0	20000	1.3229	-0.0471	1.3229	-0.0471	14000	1.43x
sgd_B1_d0	20000	1.3133	-0.0566	1.3133	-0.0566	13000	1.54x
sgd_A-fe_d1	20000	1.4559	-0.1559	1.4559	-0.1559	9000	2.22x
sgd_B0.9-fe-fresh_d1	20000	1.4601	-0.1517	1.4601	-0.1517	11500	1.74x
sgd_B1_d1	7500	1.6705	+0.0587	1.6705	+0.0587	-	-
sgd_A-fe_d2	20000	1.4051	+0.0317	1.4051	+0.0317	-	-
sgd_B1_d2	20000	1.3117	-0.0617	1.3117	-0.0617	13000	1.54x

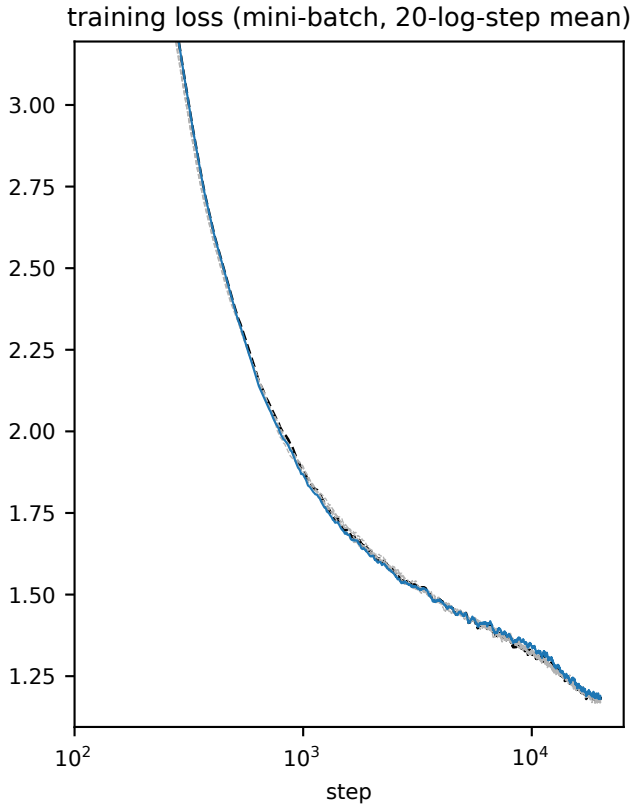

v1 SGD, TinyStories-30M: Jensen-Shannon divergence at equal training loss
data order 0
data order 1
data order 2
JS divergence (bits)
training loss (fixed training probe)
baseline: two data orders (range)
A-fe vs baseline
B0.9-fe-fresh vs baseline
B1 vs baseline
A-fe: two data orders
A-fe: two data orders
B0.9-fe-fresh: two data orders
B1: two data orders
B1: two data orders

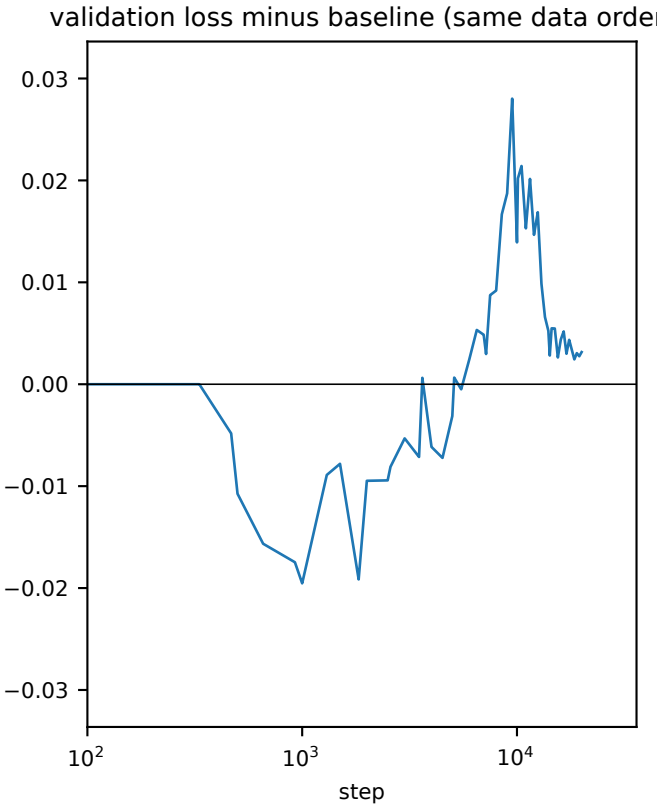

v1 SGD, TinyStories-30M: balancer trace (1)
sgd_A-fe_d0
rate / schedule (g m_a)
emb
1
2
3
4
5
6
7
8
head
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
tracked $\lambda_{max}(\hat{H})$
Lanczos check
baseline (same data order)
sgd_B0.9-fe-fresh_d0
rate / schedule (g m_a)
g (dotted)
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
sgd_B1_d0
rate / schedule (g m_a)
g (dotted)
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
sgd_A-fe_d1
rate / schedule (g m_a)
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
step
step
step


v1 SGD, TinyStories-30M: balancer trace (2)
sgd_B0.9-fe-fresh_d1
sgd_B1_d1
sgd_A-fe_d2
sgd_B1_d2
rate / schedule (g m_a)
g (dotted)
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
emb
1
2
3
4
5
6
7
8
head
tracked $\lambda_{max}(\hat{H})$
Lanczos check
baseline (same data order)
target
step

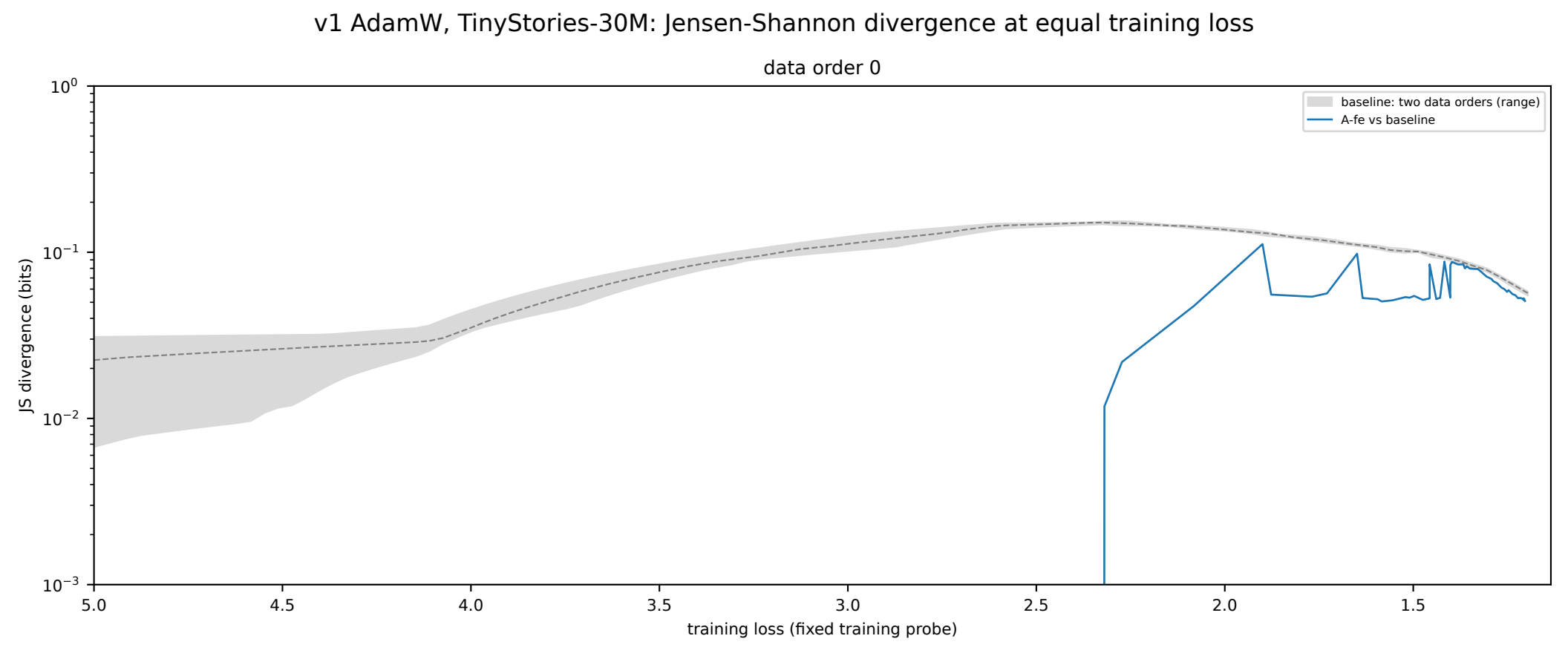
v1 AdamW, TinyStories-30M: losses

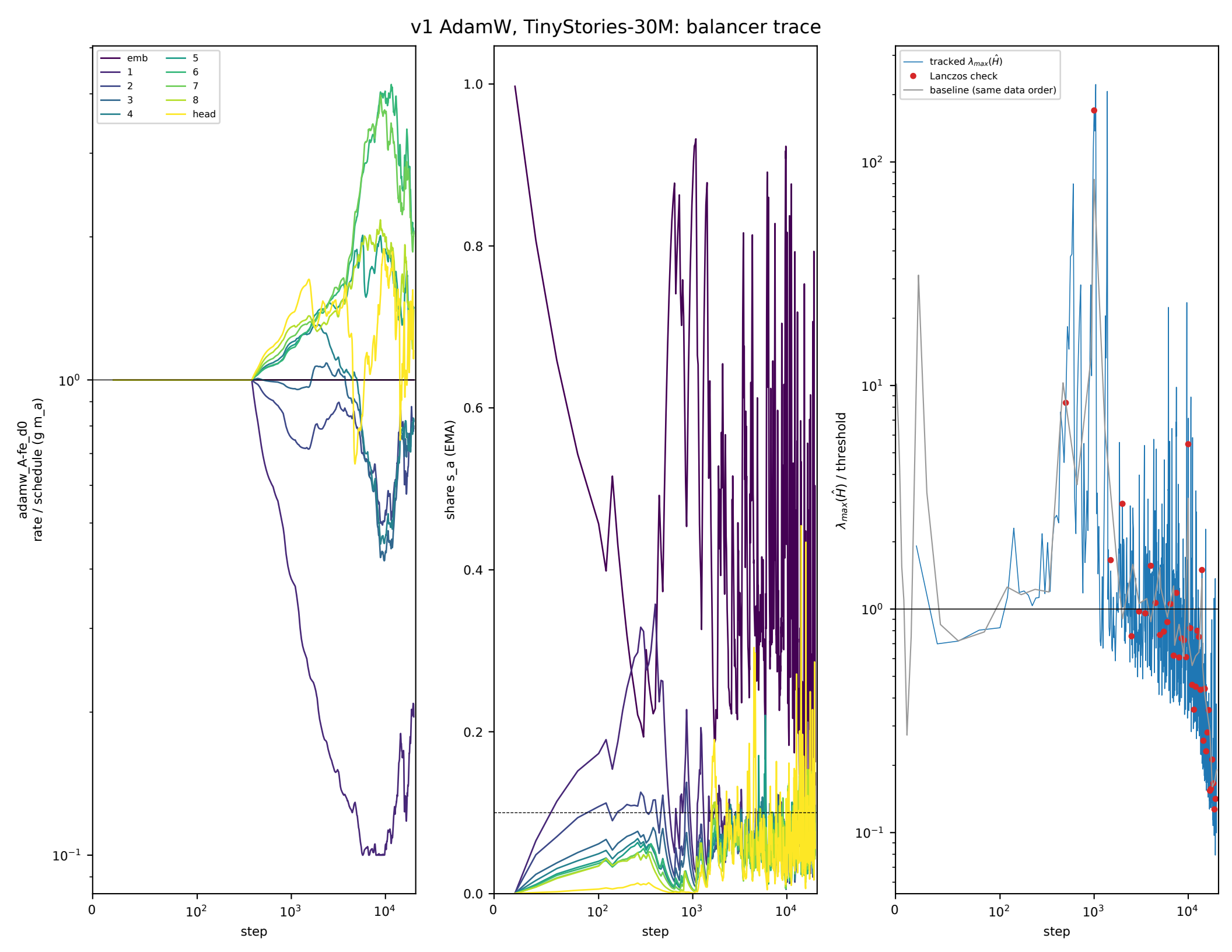




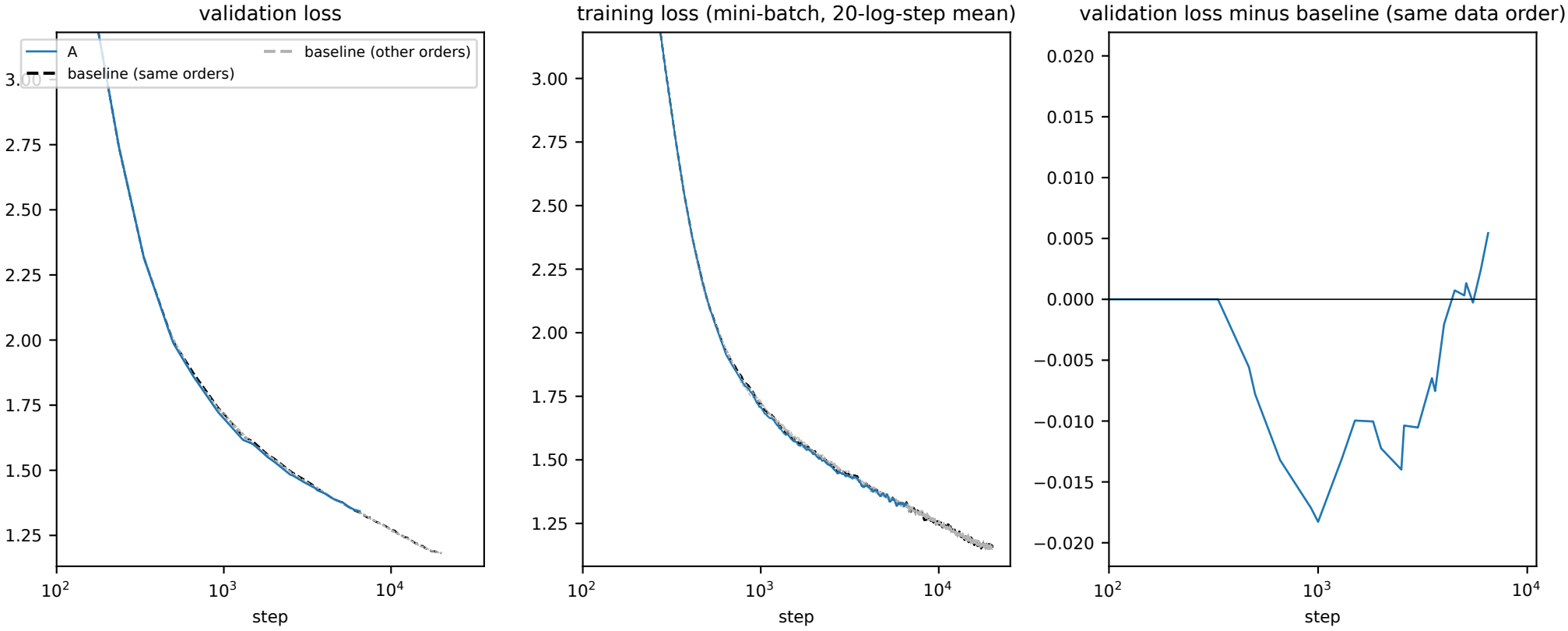


run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
adamw_A-fe_d0	20000	1.2040	+0.0032	1.2040	+0.0032	-	-

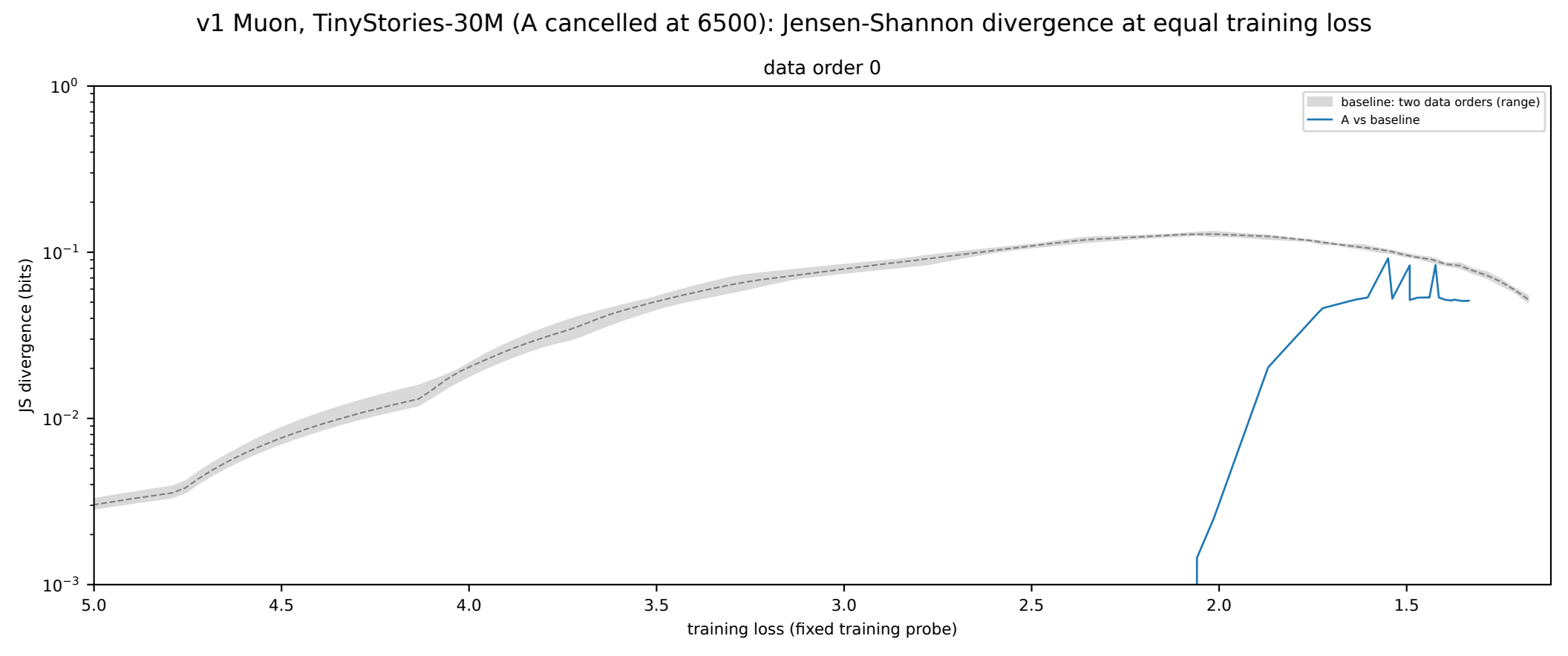

v1 AdamW, TinyStories-30M: Jensen-Shannon divergence at equal training loss
data order 0
baseline: two data orders (range)
A-fe vs baseline
JS divergence (bits)
10^{0}
10^{-1}
10^{-2}
10^{-3}
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
training loss (fixed training probe)

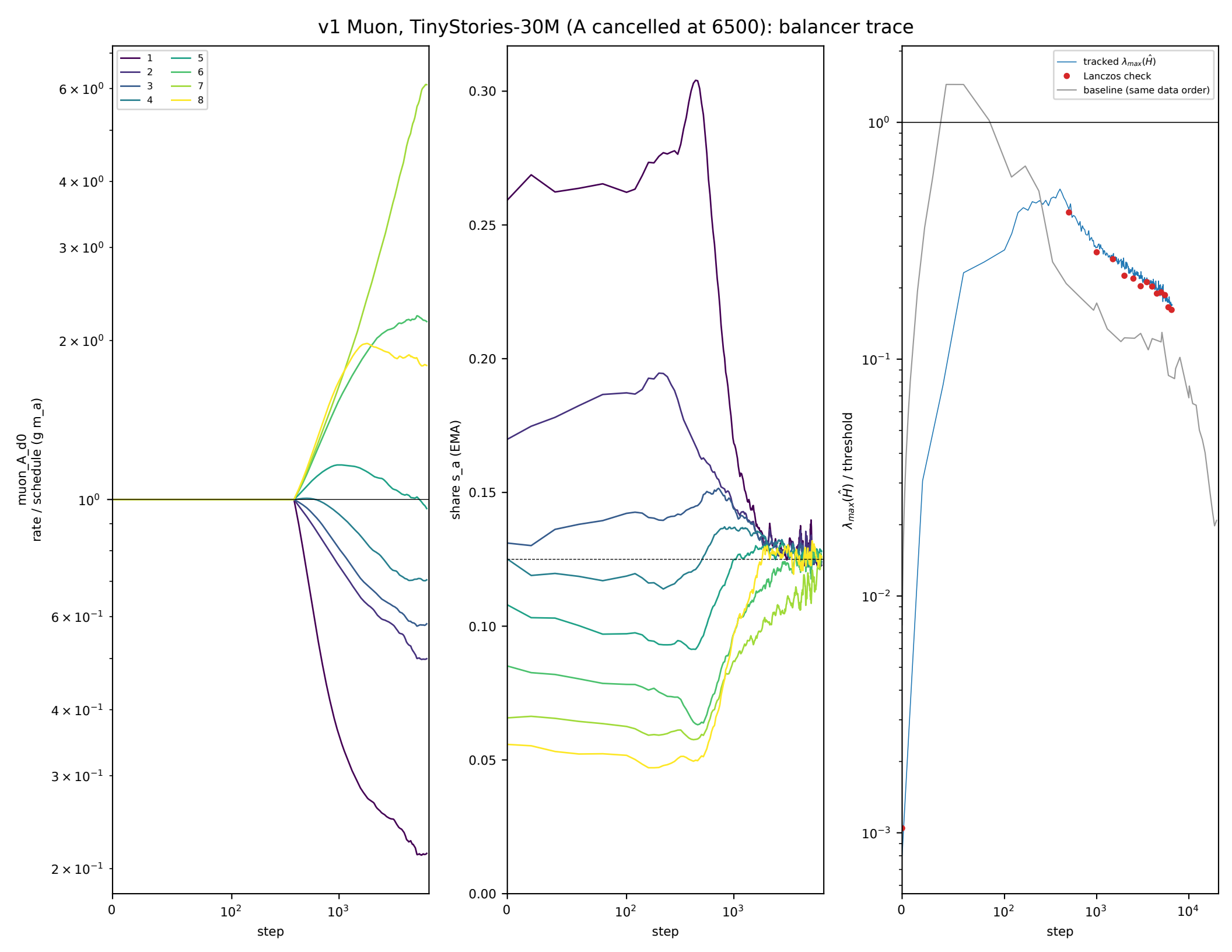

v1 AdamW, TinyStories-30M: balancer trace
emb
1
2
3
4
5
6
7
8
head
adamw_A-fe_d0
rate / schedule (g m_a)
share s_a (EMA)
tracked $\lambda_{max}(\hat{H})$
Lanczos check
baseline (same data order)
$\lambda_{max}(\hat{H})$ / threshold
step
step
step

v1 Muon, TinyStories-30M (A cancelled at 6500): losses

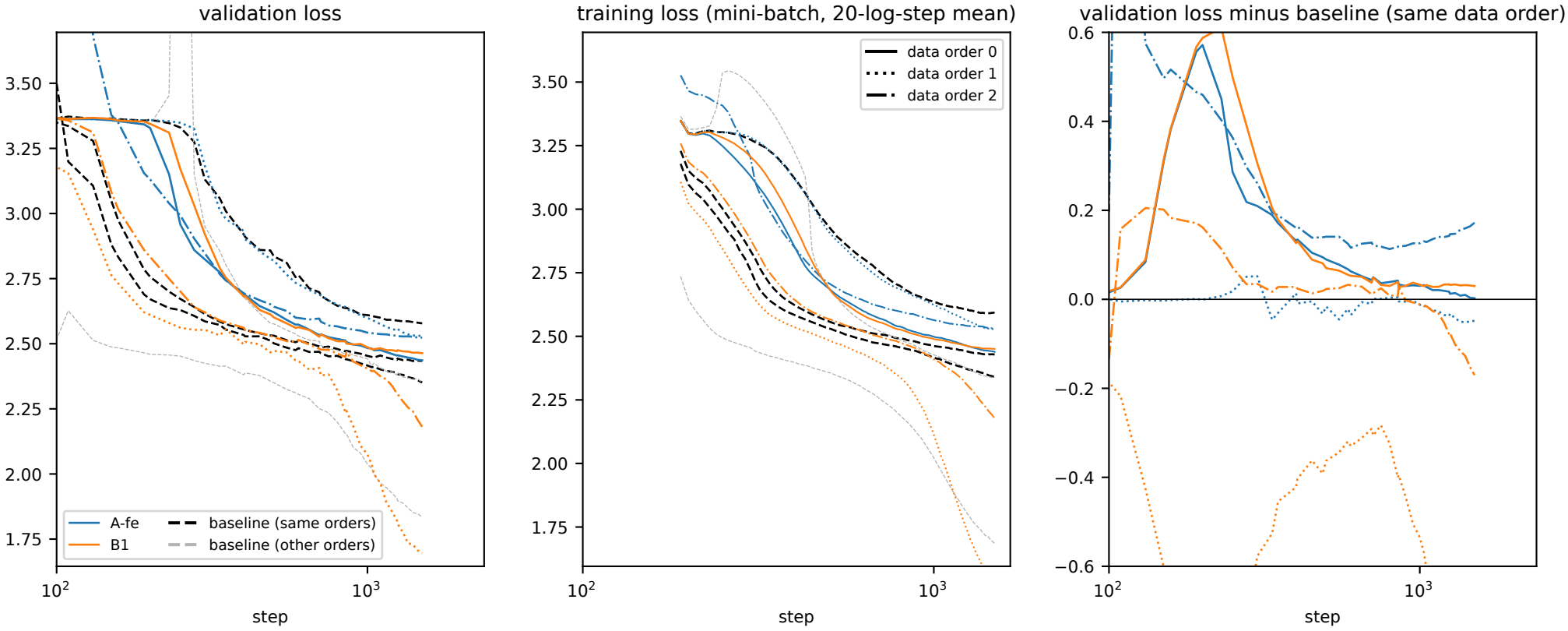


run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
muon_A_d0	6500	1.3429	+0.1616	1.3429	+0.1616	-	-

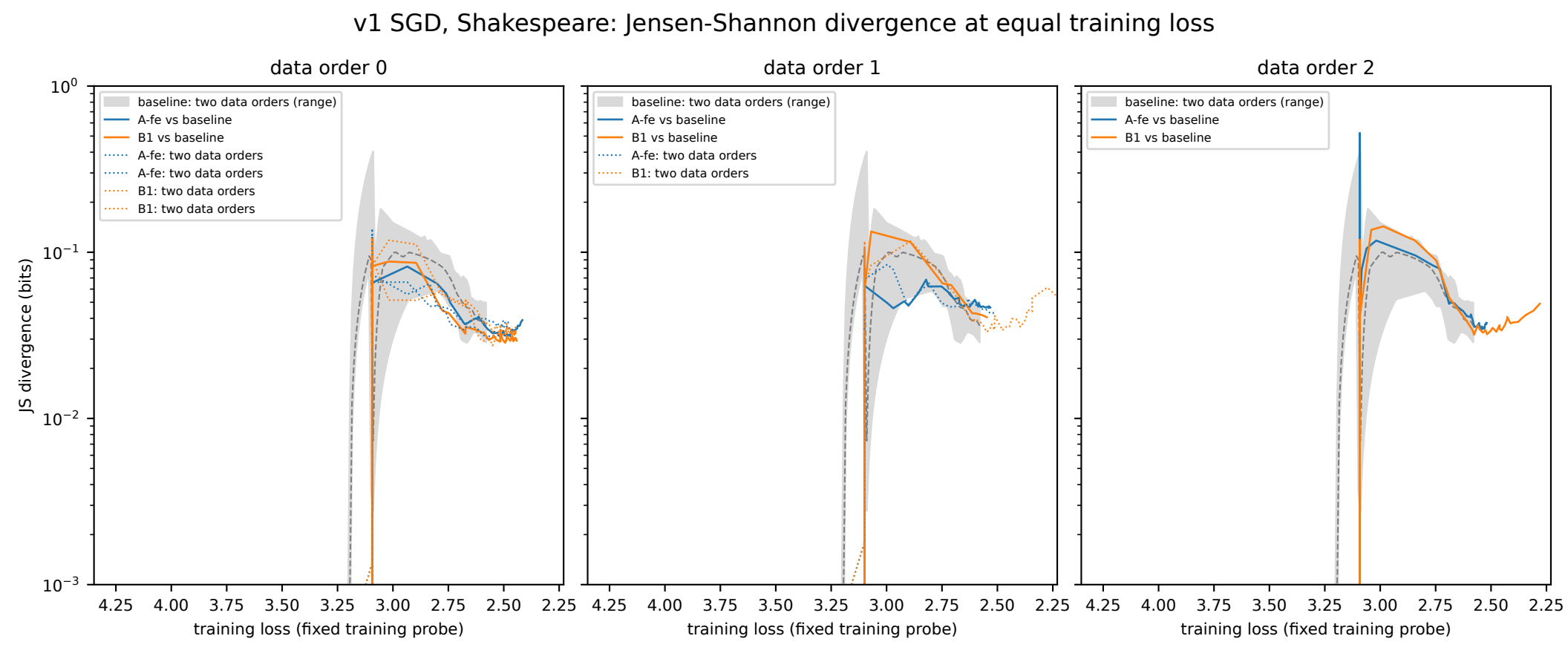

v1 Muon, TinyStories-30M (A cancelled at 6500): Jensen-Shannon divergence at equal training loss
data order 0
baseline: two data orders (range)
A vs baseline
JS divergence (bits)
training loss (fixed training probe)
10^0
10^-1
10^-2
10^-3
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5


v1 Muon, TinyStories-30M (A cancelled at 6500): balancer trace
muon_A_d0
rate / schedule (g m_a)
step
share s_a (EMA)
step
$\lambda_{max}(\hat{H})$ / threshold
step
tracked $\lambda_{max}(\hat{H})$
Lanczos check
baseline (same data order)
1
2
3
4
5
6
7
8

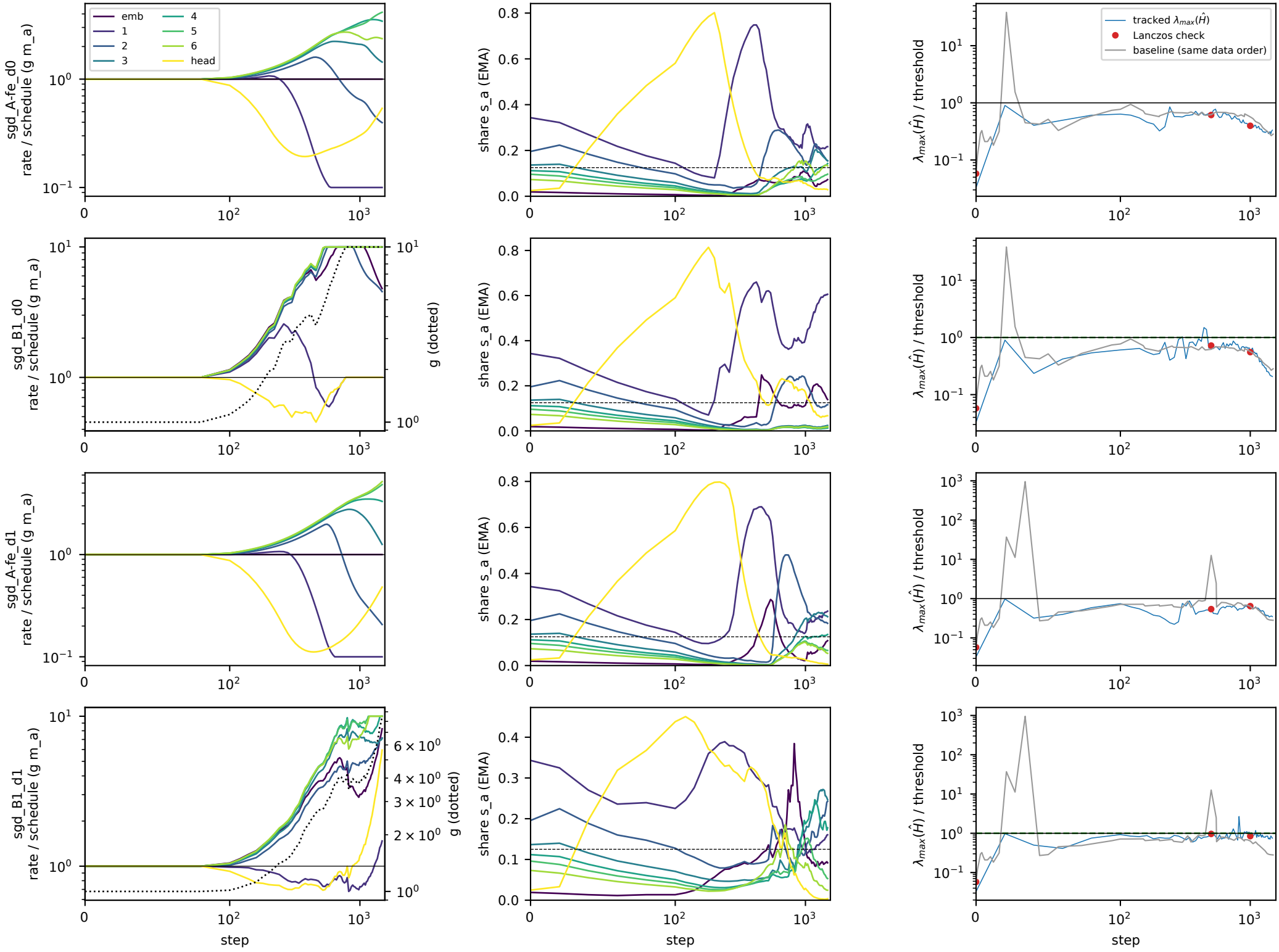
v1 SGD, Shakespeare: losses



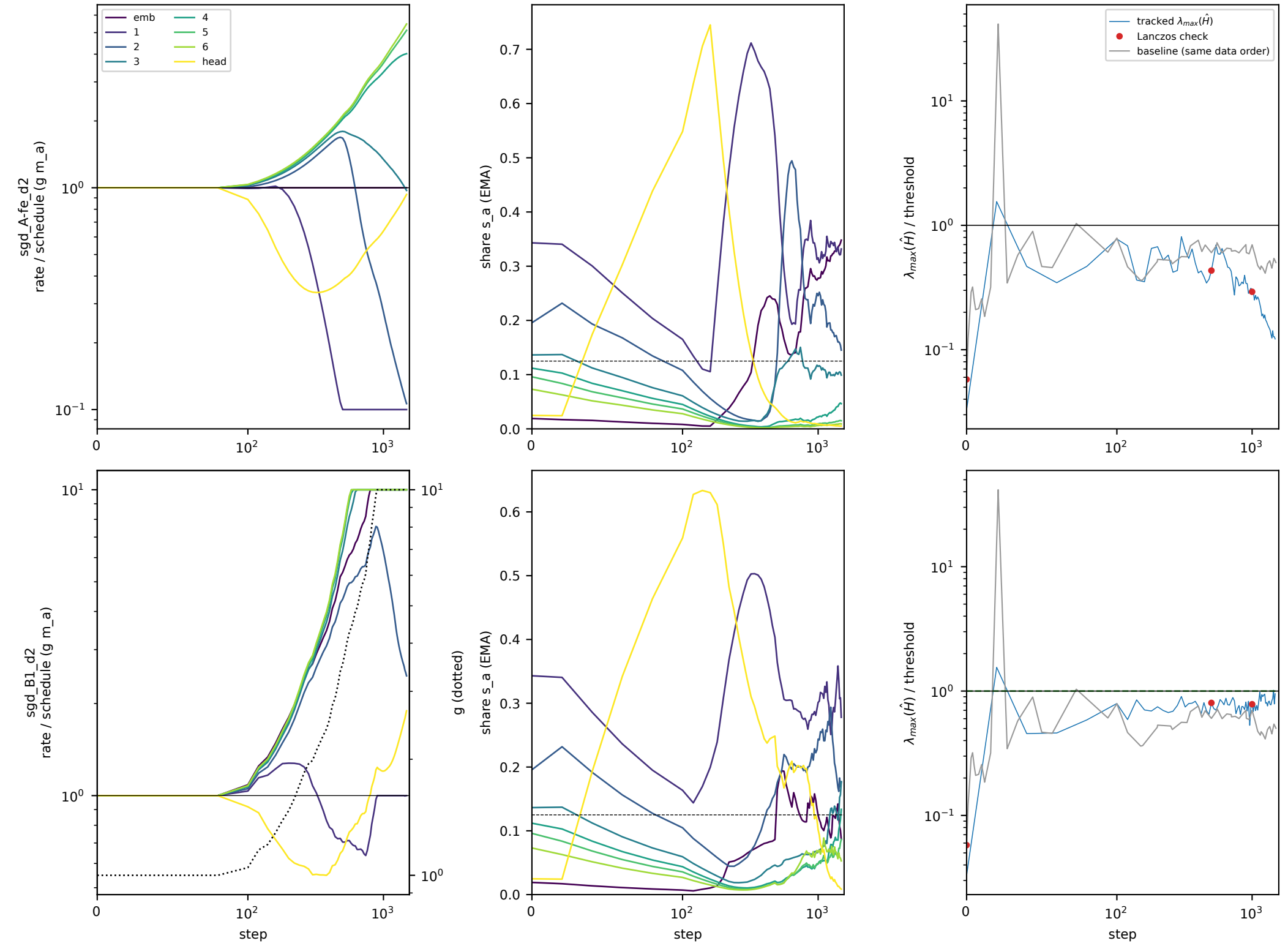
run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
sgd_A-fe_d0	1500	2.4363	+0.0024	2.4363	+0.0027	-	-
sgd_B1_d0	1500	2.4637	+0.0298	2.4637	+0.0301	-	-
sgd_A-fe_d1	1500	2.5300	-0.0481	2.5300	-0.0481	1100	1.36x
sgd_B1_d1	1500	1.6955	-0.8826	1.6955	-0.8826	250	6.00x
sgd_A-fe_d2	1500	2.5234	+0.1721	2.5234	+0.1721	-	-
sgd_B1_d2	1500	2.1810	-0.1703	2.1810	-0.1703	1200	1.25x


v1 SGD, Shakespeare: Jensen-Shannon divergence at equal training loss
data order 0
data order 1
data order 2
JS divergence (bits)
training loss (fixed training probe)
baseline: two data orders (range)
A-fe vs baseline
B1 vs baseline
A-fe: two data orders
A-fe: two data orders
B1: two data orders
B1: two data orders

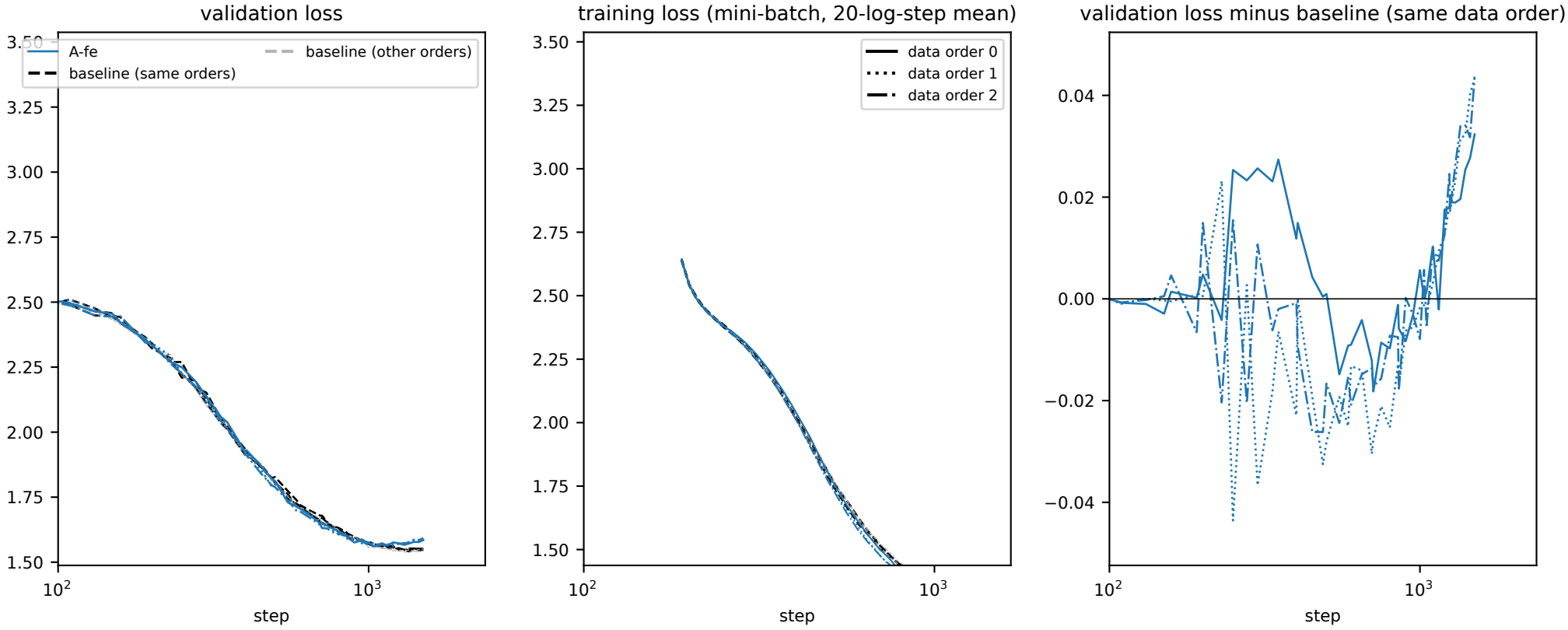
v1 SGD, Shakespeare: balancer trace (1)



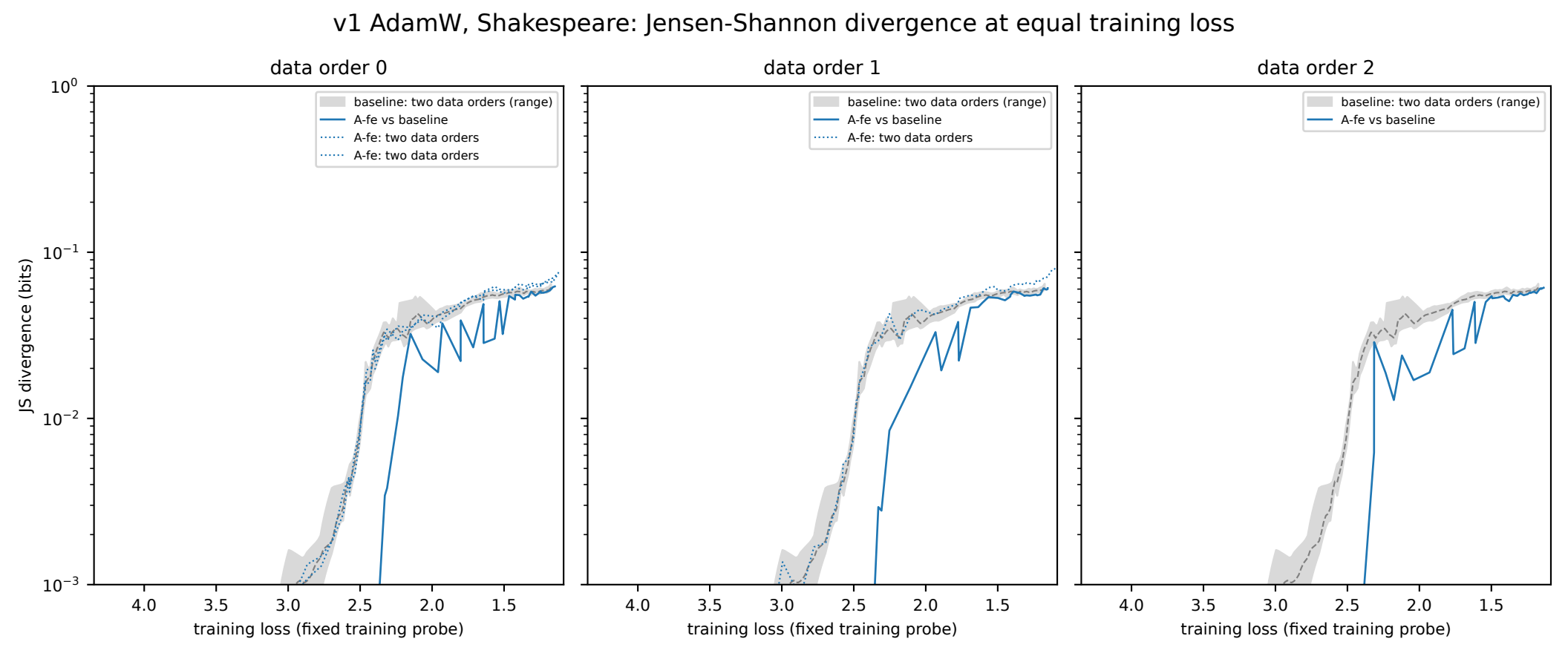
v1 SGD, Shakespeare: balancer trace (2)

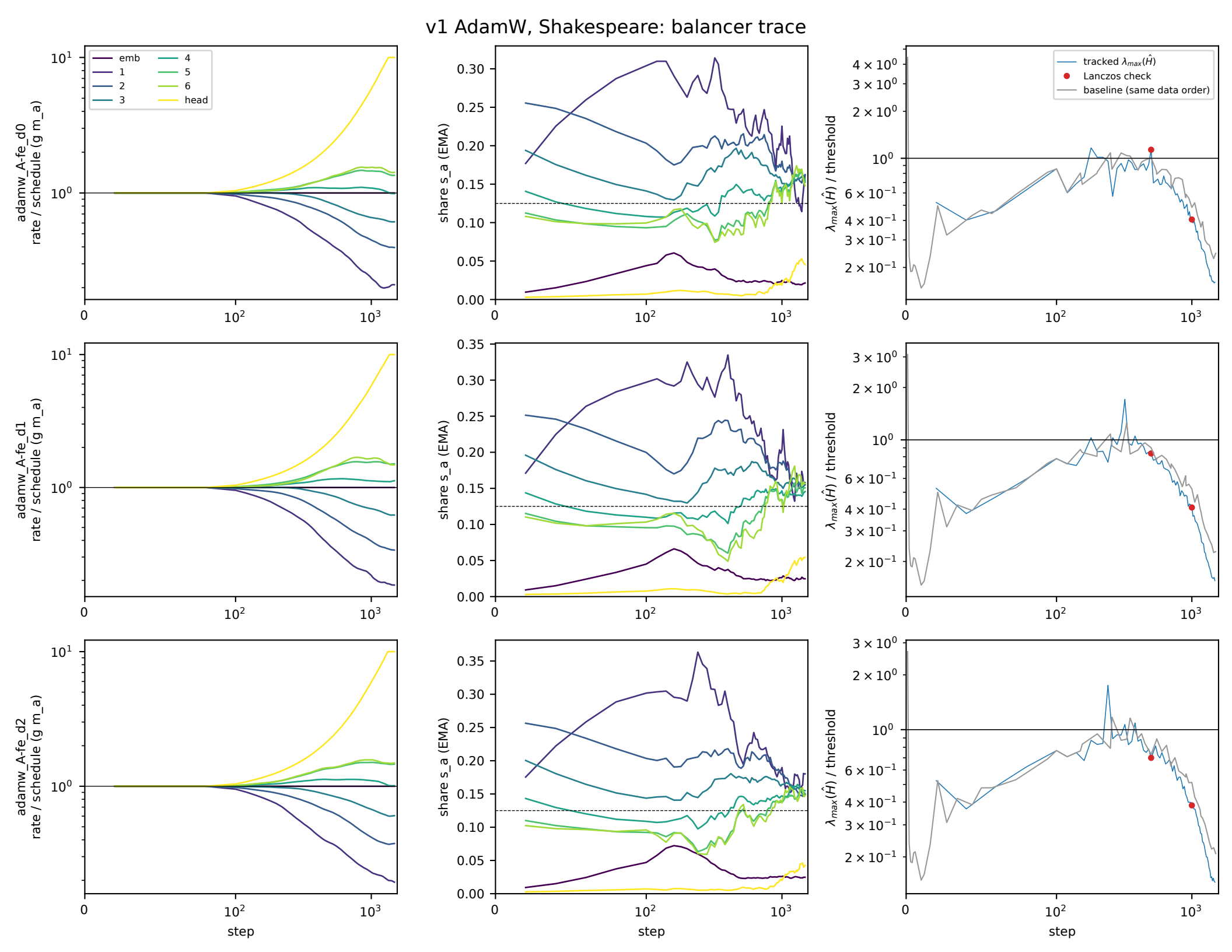


v1 AdamW, Shakespeare: losses

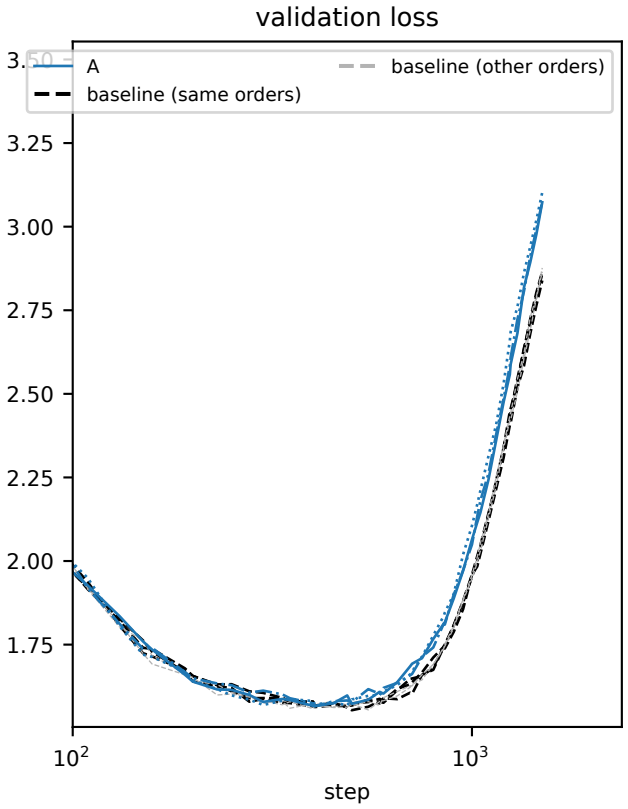


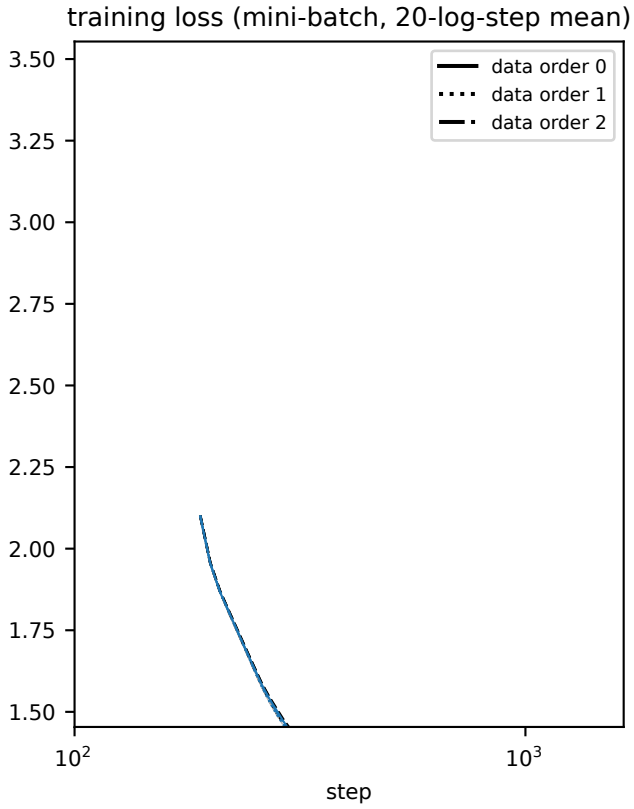
run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
adamw_A-fe_d0	1500	1.5848	+0.0323	1.5605	+0.0129	-	-
adamw_A-fe_d1	1500	1.5899	+0.0434	1.5627	+0.0171	-	-
adamw_A-fe_d2	1500	1.5927	+0.0437	1.5633	+0.0216	-	-

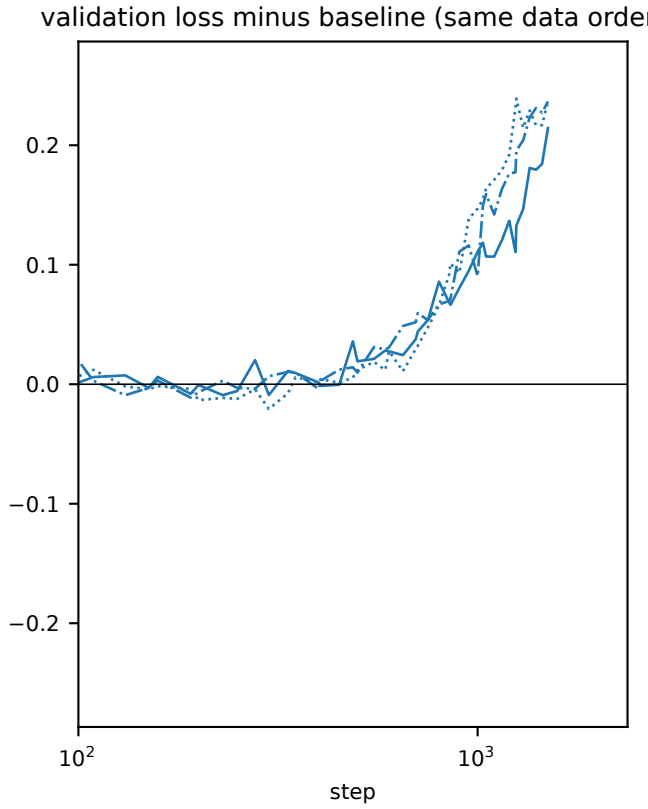

v1 AdamW, Shakespeare: Jensen-Shannon divergence at equal training loss
data order 0
data order 1
data order 2
JS divergence (bits)
10^{0}
10^{-1}
10^{-2}
10^{-3}
baseline: two data orders (range)
A-fe vs baseline
A-fe: two data orders
A-fe: two data orders
4.0
3.5
3.0
2.5
2.0
1.5
training loss (fixed training probe)


v1 AdamW, Shakespeare: balancer trace
adamw_A-fe_d0
rate / schedule (g m_a)
adamw_A-fe_d1
rate / schedule (g m_a)
adamw_A-fe_d2
rate / schedule (g m_a)
emb
1
2
3
4
5
6
head
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
tracked $\lambda_{max}(\hat{H})$
Lanczos check
baseline (same data order)
step

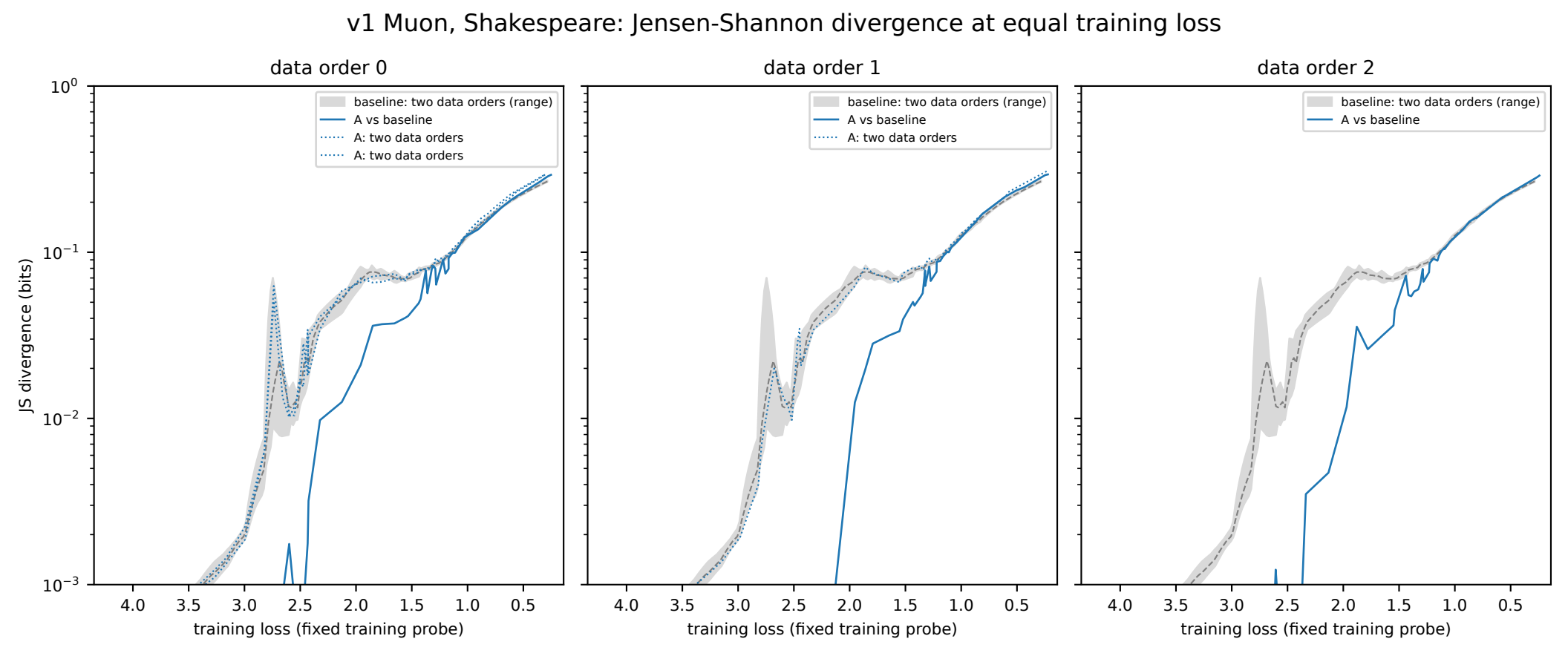
v1 Muon, Shakespeare: losses

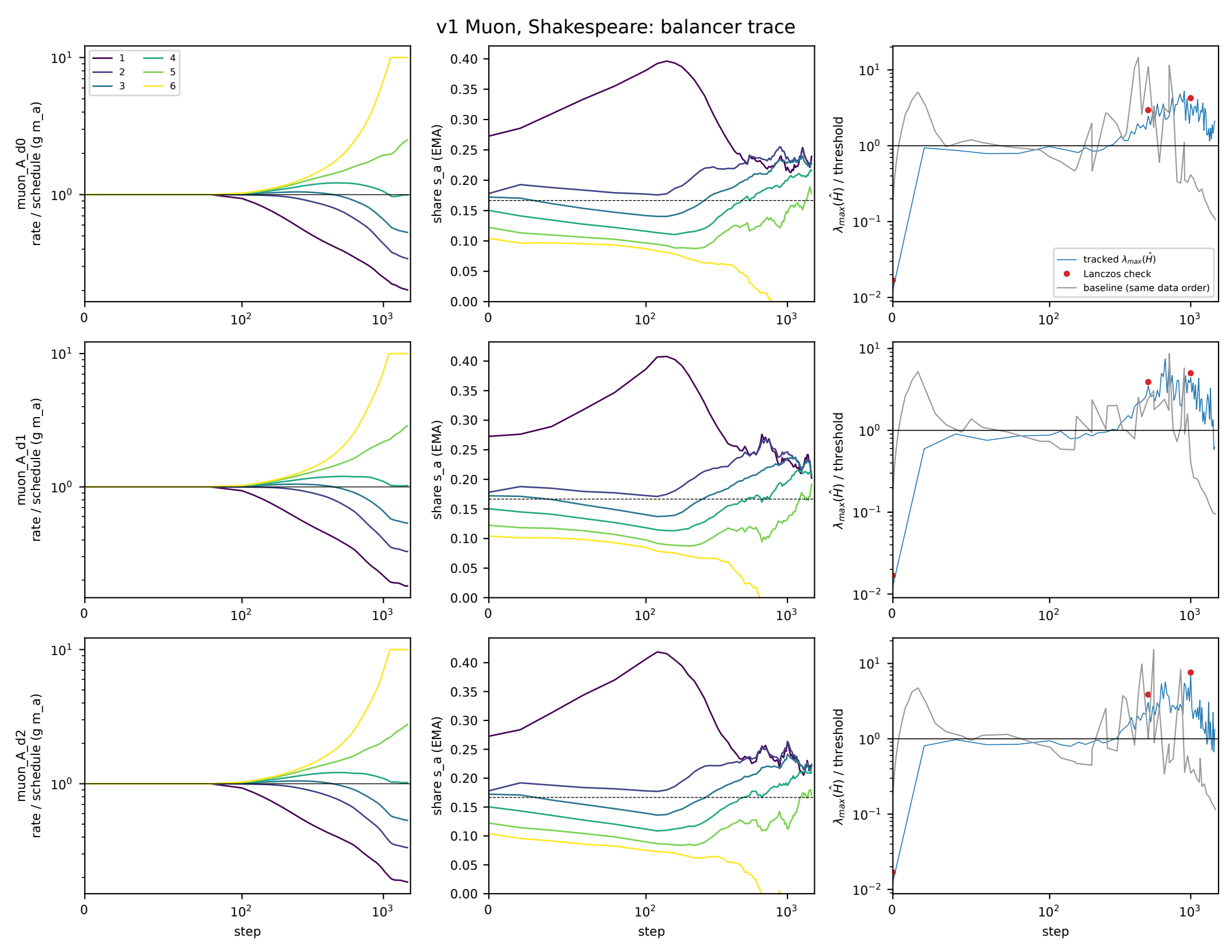




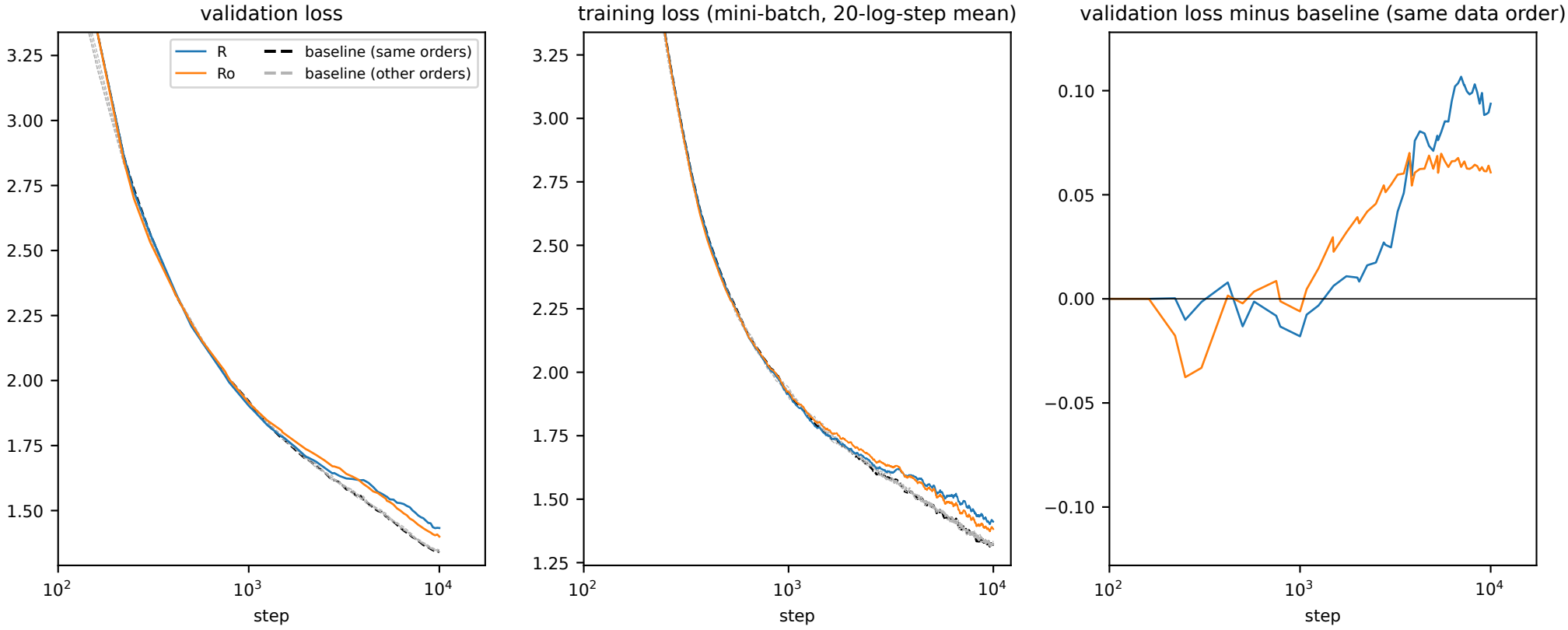


run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
muon_A_d0	1500	3.0729	+0.2143	1.5660	+0.0123	-	-
muon_A_d1	1500	3.1027	+0.2370	1.5684	+0.0055	-	-
muon_A_d2	1500	3.0756	+0.2366	1.5661	+0.0028	-	-

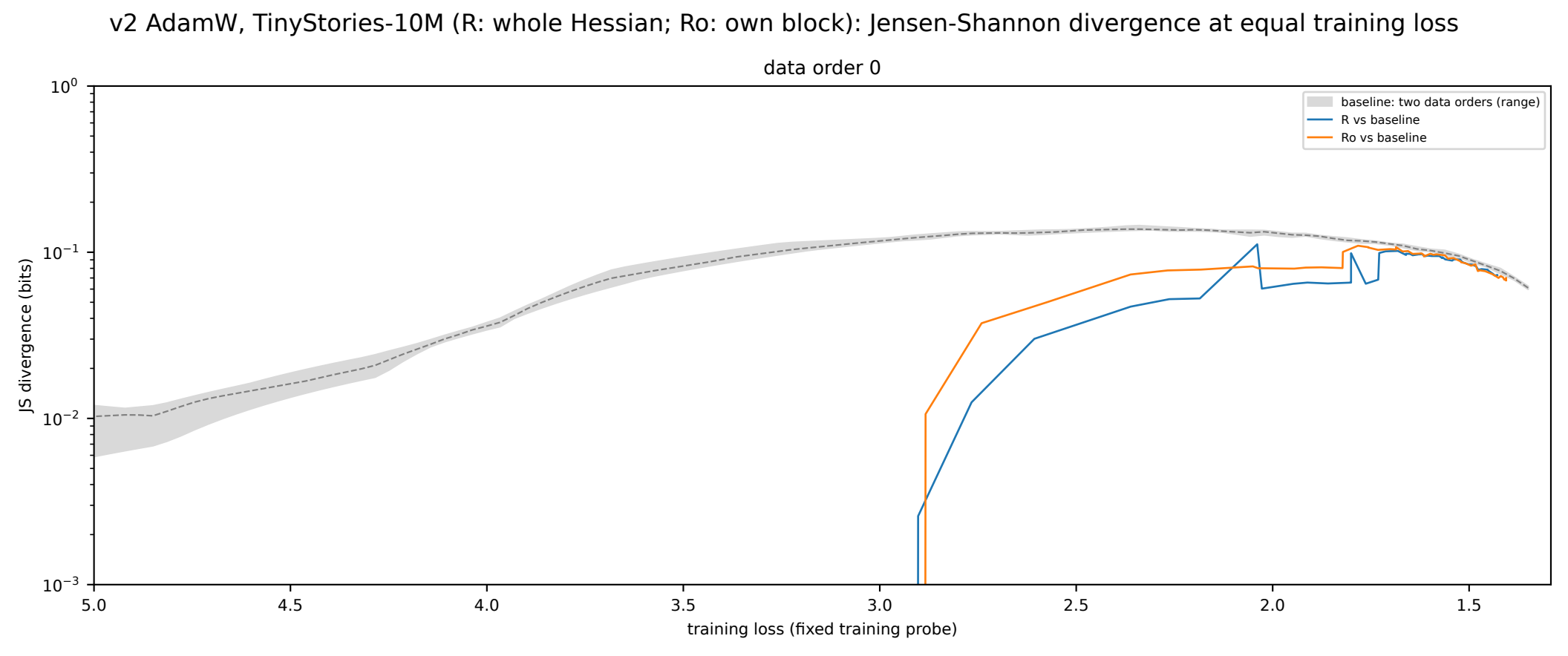

v1 Muon, Shakespeare: Jensen-Shannon divergence at equal training loss
data order 0
data order 1
data order 2
JS divergence (bits)
training loss (fixed training probe)
baseline: two data orders (range)
A vs baseline
A: two data orders
A: two data orders

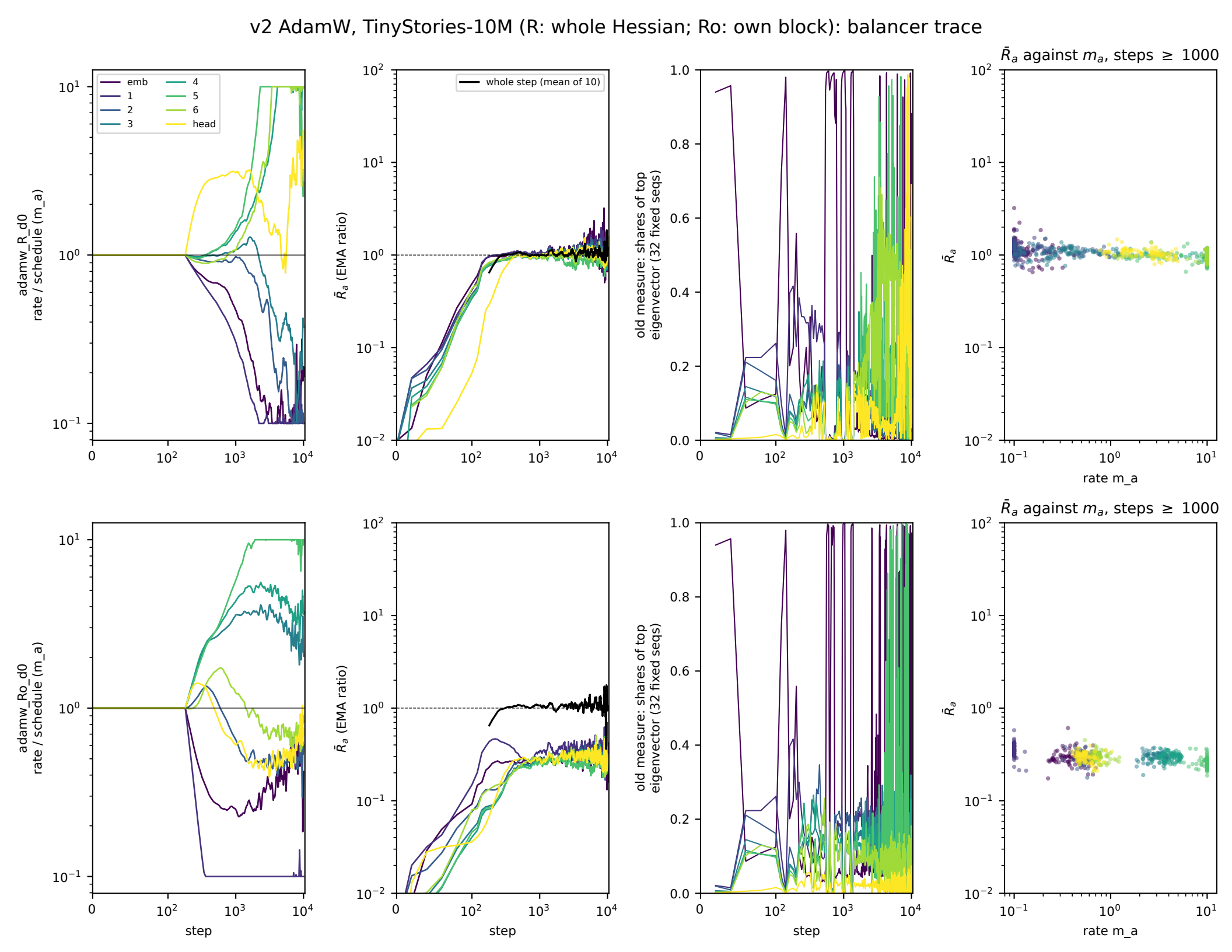

v1 Muon, Shakespeare: balancer trace
muon_A_d0
rate / schedule (g m_a)
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
muon_A_d1
rate / schedule (g m_a)
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
muon_A_d2
rate / schedule (g m_a)
share s_a (EMA)
$\lambda_{max}(\hat{H})$ / threshold
1
2
3
4
5
6
tracked $\lambda_{max}(\hat{H})$
Lanczos check
baseline (same data order)
step
step

v2 AdamW, TinyStories-10M (R: whole Hessian; Ro: own block): losses

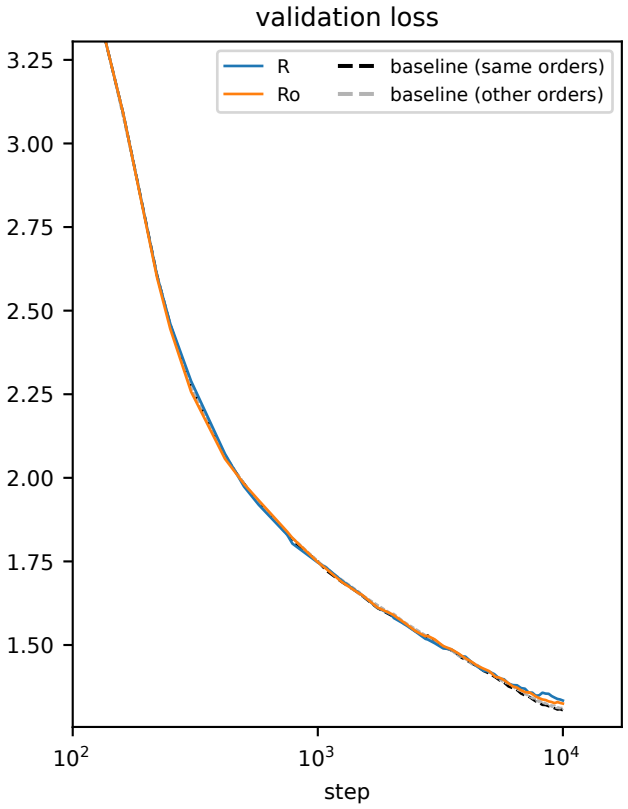


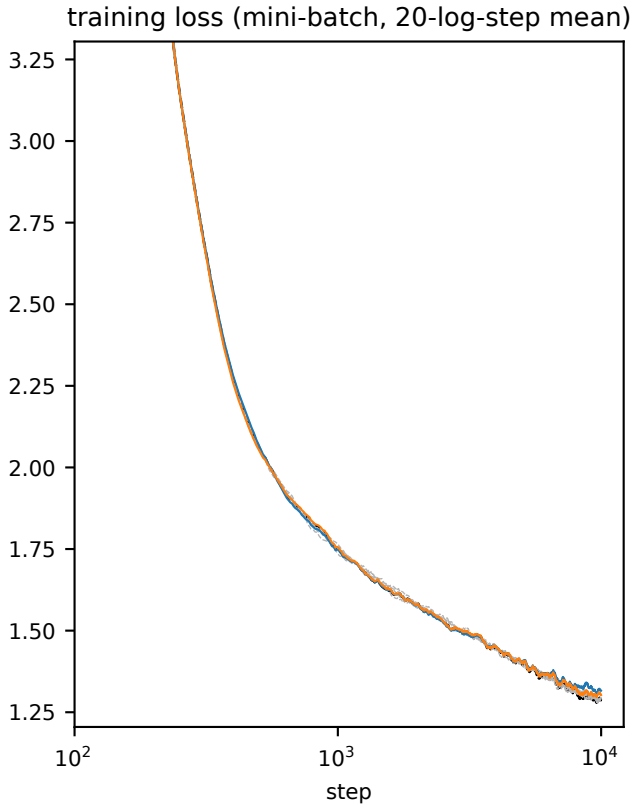
run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
adamw_R_d0	10000	1.4329	+0.0937	1.4323	+0.0932	-	-
adamw_Ro_d0	10000	1.3999	+0.0607	1.3999	+0.0607	-	-

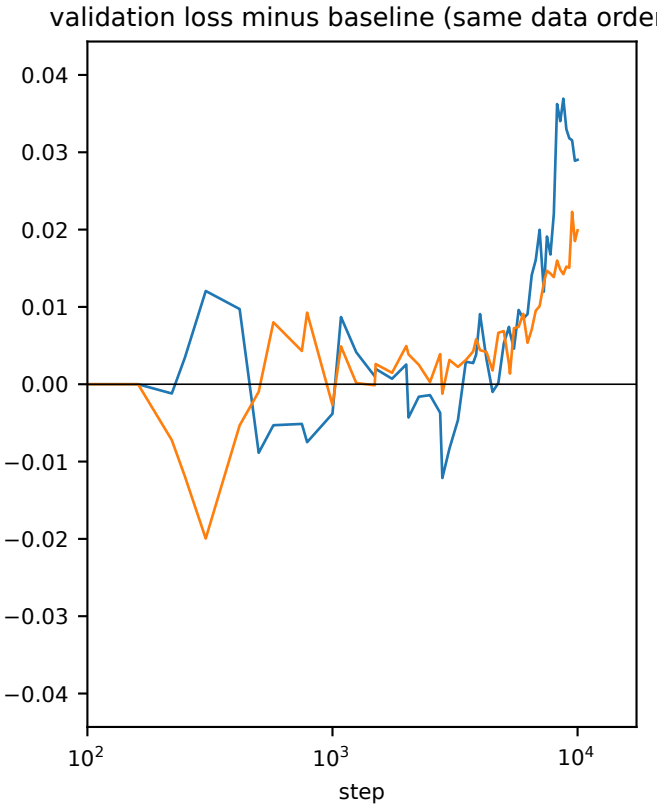

v2 AdamW, TinyStories-10M (R: whole Hessian; Ro: own block): Jensen-Shannon divergence at equal training loss
data order 0
baseline: two data orders (range)
R vs baseline
Ro vs baseline
JS divergence (bits)
training loss (fixed training probe)
10^0
10^-1
10^-2
10^-3
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5


v2 AdamW, TinyStories-10M (R: whole Hessian; Ro: own block): balancer trace
adamw_R_d0
rate / schedule (m_a)
emb
1
2
3
4
5
6
head
$\bar{R}_a$ (EMA ratio)
whole step (mean of 10)
old measure: shares of top
eigenvector (32 fixed seqs)
$\bar{R}_a$ against m_a, steps ≥ 1000
$\bar{R}_a$
rate m_a
adamw_Ro_d0
rate / schedule (m_a)
$\bar{R}_a$ (EMA ratio)
old measure: shares of top
eigenvector (32 fixed seqs)
$\bar{R}_a$ against m_a, steps ≥ 1000
$\bar{R}_a$
rate m_a
step
step
step

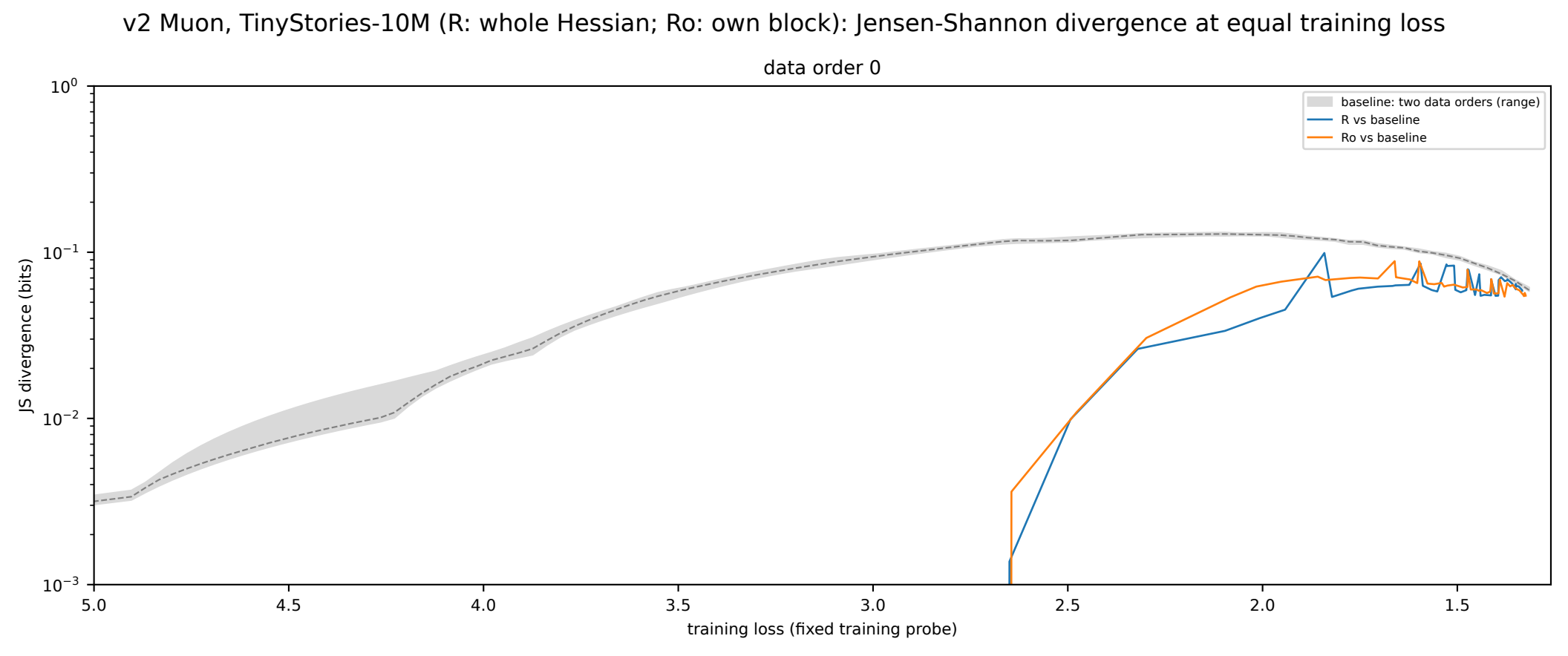
v2 Muon, TinyStories-10M (R: whole Hessian; Ro: own block): losses

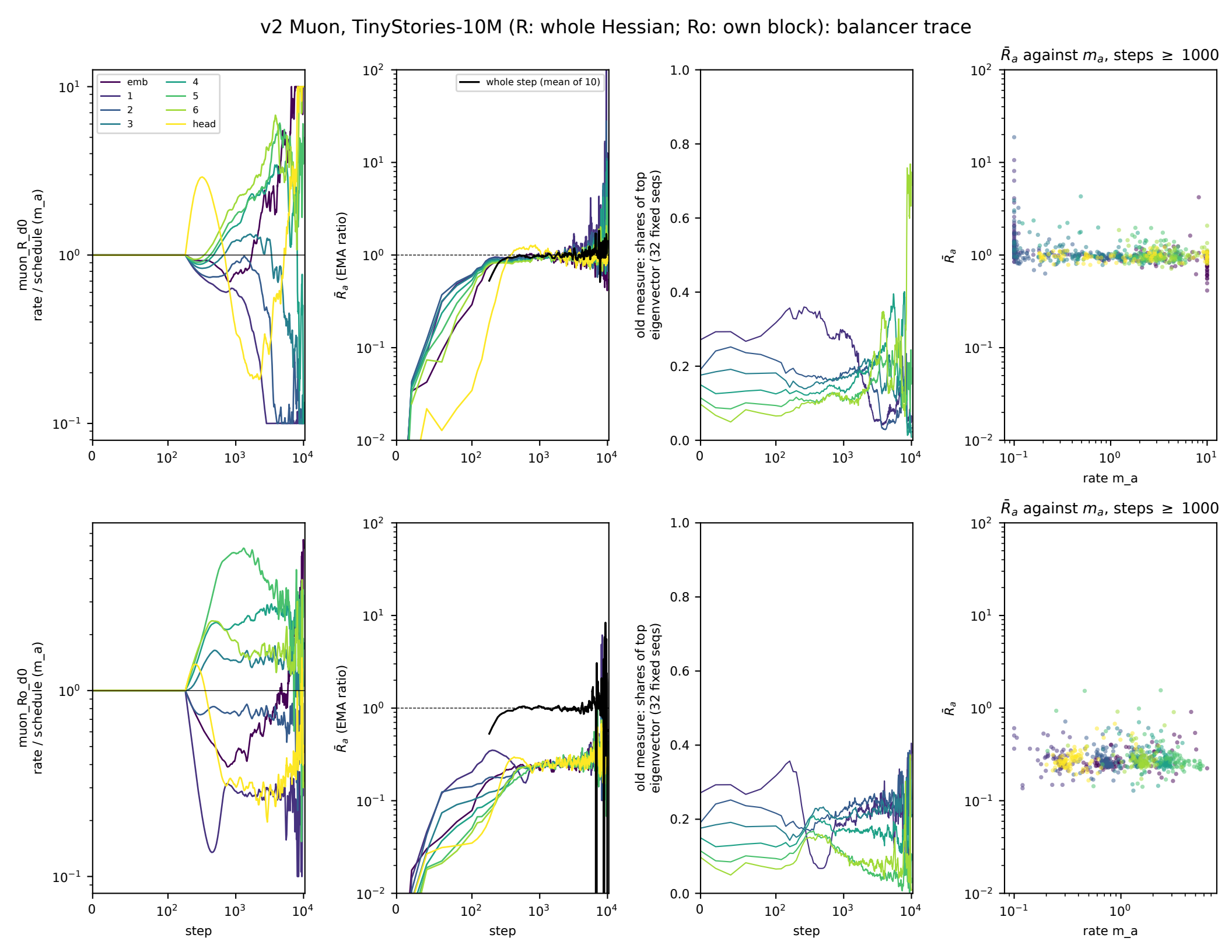






run	last step	final val	minus baseline	best val	minus baseline	reaches baseline best	speedup
muon_R_d0	10000	1.3339	+0.0290	1.3339	+0.0290	-	-
muon_Ro_d0	10000	1.3248	+0.0199	1.3248	+0.0199	-	-


v2 Muon, TinyStories-10M (R: whole Hessian; Ro: own block): Jensen-Shannon divergence at equal training loss
data order 0
baseline: two data orders (range)
R vs baseline
Ro vs baseline
JS divergence (bits)
training loss (fixed training probe)
10^0
10^-1
10^-2
10^-3
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5


v2 Muon, TinyStories-10M (R: whole Hessian; Ro: own block): balancer trace
muon_R_d0
rate / schedule (m_a)
emb
1
2
3
4
5
6
head
$\bar{R}_a$ (EMA ratio)
whole step (mean of 10)
old measure: shares of top
eigenvector (32 fixed seqs)
$\bar{R}_a$ against m_a, steps $\geq$ 1000
$\bar{R}_a$
rate m_a
muon_Ro_d0
rate / schedule (m_a)
step